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Identifiability and identification of inertial parameters using the underactuated base-link dynamics for legged multibody systems

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Abstract

In this paper we study the dynamics of multibody systems with the base not permanently fixed to the inertial frame, or more specifically legged systems such as humanoid robots and humans. The issue is to be approached in terms of the identification theory developed in the field of robotics. The under-actuated base-link which characterizes the dynamics of legged systems is the focus of this work. The useful mechanical feature to analyze the dynamics of legged system is proven: the set of inertial parameters appearing in the equation of motion of the under-actuated base is equivalent to the set in the equations of the whole body. In particular, when no external force acts on the system, all of the parameters in the set except the total mass are generally identifiable only from the observation of the free-flying motion. We also propose a method to identify the inertial parameters based on the dynamics of the under-actuated base. The method does not require the measurement of the joint torques. Neither the joint frictions nor the actuator dynamics need to be considered. Even when the system has no external reaction force, the method is still applicable. The method has been tested on both a humanoid robot and a human, and the experimental results are shown.

Keywords

Humanoid robots, free-floating systems, inertial parameters, identification, identifiability

1. Introduction

1.1. Demand on mass-property identification of humanoid robots and humans

Similarity of appearance between robots and humans would be appreciated when the robots work alongside humans in the daily human environments or when the robots work alone in the human environments in the absence of humans. Under such expectations, many kinds of humanoid robots have been developed and studied since the early 1970s (Kato, 1973; Hirai et al., 1998; Kaneko et al., 2004; Cheng et al., 2006). Nevertheless, many humanoids still lack the fluency of motion. There are many unsolved problems in motion generation, planning, and control. The stability of biped motion and the safety of interaction with humans are typical control problems. The fast and precise motion control of a robot generally requires an accurate model of dynamics. The geometric parameters such as joint location can be easily obtained from the CAD data. The inertial parameters such as mass, the center of mass (COM), and inertia tensor of each link can be estimated as well. However, the identification of inertial parameters by experiments is still necessary, because the body of robots is not in detail a set of separate links connected at the joints. It is not clear which part of mechanical transmissions such as motors, gears, and shafts belongs to each link. It becomes even more unclear when we need to consider wires for electric circuits, whose mass is not negligible. Needless to say, the identification is also useful when a humanoid robot is equipped with an additional tool or carries a backpack.

The analysis of human motions provides a clue to designing the control system of humanoid robots. On the other hand, the biomechanical analysis of human movements is useful for the study of motor control of humans. Estimation of muscle activities from the observed human motion has been studied by Delp and Loan (1995), Yamane and Nakamura (2000), and Nakamura et al. (2005). The human biomechanics studies find applications in sports training, healthcare, and medical diagnosis/treatments for musculoskeletal disorders in orthopedics, cardiology, and neurology. It is obvious that the biomechanical modeling of a

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human body depends on the precision of the inertial properties of a human body. The identification is in high demand in the field, because the literature data of inertia parameters is commonly used without taking account of differences between patients or subjects.

1.2. Inertial parameters identification of multibody systems

System identification is, in general, the problem of building a mathematical model of a targeted system from its measured data. In the field of robotics, the identification of inertial parameters of multibody systems has been developed as the original theory. Although the inertial parameters of multibody systems seem to exist indistinctly in the equations of motion, they actually appear in a linear form and thus can be separated linearly from the geometric parameters and the generalized coordinates (Mayeda et al., 1984; Atkeson et al., 1986). Assuming that the geometric parameters and the coordinates are known, the unknown inertial parameters can be identified from the measured data by making use of the linearity.

It is also known that the inertial parameters are redundant when describing the equations of motion and, therefore, they cannot be identified uniquely from the data of link coordinates, joint torques, and external forces. In other words, the inertial parameters of robots are not structurally identifiable (Bellman and Aström, 1970) from the equations of motion. Only the minimal set of inertial parameters that describes the dynamics of the system can be identified. This minimal set is called the base parameters (Mayeda et al., 1990; Gautier, 1990; Kawasaki et al., 1991; Khalil and Bennis, 1995). It is computed from the inertial parameters by eliminating those that have no influence on the model and regrouping some according to the kinematics of the system. Symbolical or numerical computation methods of base parameters for general robots are now available.

The least-squares method is often used to identify base parameters because of its ease of implementation (Khalil and Dombre, 2002). However, base parameters cannot be identified if improper data of motion is used for identification. The base parameters are only guaranteed to be identified under ideal conditions. The motions that excite the system dynamics and satisfies *a posteriori* identifiability of the base parameters is called persistent exciting (PE) trajectories or optimal excitation trajectories. To improve the performance of identification, the methods to design the PE trajectories of robot manipulators have been studied and developed (Gautier and Khalil, 1992; Swevers et al., 1997; Park, 2006).

Although the general formalisms and methods are now available (Khalil and Dombre, 2002; Sujan and Dubowsky, 2003; Khalil et al., 2007; Nakanishi et al., 2008) and have been applied to systems in other fields (Murotsu et al., 1994; Yoshida and Abiko, 2002; Venture et al., 2009b), the applications to mobile robotics (Venture et al., 2006), in particular, to legged robots (Ayusawa et al., 2008; Venture et al., 2008), are still sparse. The main problems in identifying the parameters of legged systems such as humanoid robots and humans are explained as follows. Most common identification methods require the measurement of the generalized coordinates, each joint torque, and the external forces acted on the system. Joint torque can either be measured directly by a torque sensor or calculated from the actuator models. The actuator models include transmissions, which commonly suffer from large friction due to gears. Therefore, the accuracy of inertial parameters is limited by the difficulty of modeling frictions. Moreover, many humanoid robots are rarely equipped with torque sensors in their limbs because of their large number of degrees of freedom (DOFs) and the constraints of size and weight. In particular, it is of great difficulty to measure the joint torque of a human during movement.

1.3. Dynamics study of a legged mechanism

Dynamics of legged systems is characterized by the absence of a link fixed with respect to the inertial frame. Legged systems, which control reaction force by leg manipulation so as to move, have the same mechanical features as free-flying systems such as free-floating space robots. If we represent a legged system as a kinematic tree structure in the same way as a robot manipulator, the generalized input forces which actuate the 6 DOFs of the root of the structure, or the base-link, are always equal to zero. Any link of a legged system can be chosen as the base-link; if another link is chosen as a new base-link, the base of the tree structure is to be changed to the new one, and the new one is to be subject to the constraint of its generalized input forces instead. Owing to the constraint, the generalized external forces of the equations of motion corresponding to the base-link are equivalent to the total forces and moments acting on the system.

In general, the derivation of the low-dimensional model of complex nonlinear systems also facilitates their analysis. In the control theory of biped robots, the inverted pendulum model is often used (Miyazaki and Arimoto, 1980; Furusho and Masubuchi, 1986). For legged locomotion, the total COM and zero moment point (ZMP) (Vukobratovic and Stepanenko, 1972) are also fundamental to analyze and design motion patterns. Recently, the inverted pendulum model, which assumes that the total mass is concentrated at the total COM and is driven by ZMP manipulation, shows its usefulness for control of biped systems (Sugihara et al., 2002; Kajita et al., 2001). This is because the model is equivalent to the equations of motion of the base-link concerning translational movement, which represent the core dynamics of legged systems. In such models, the dynamics of inertia tensor is often eliminated. Some humanoid studies use not only the COM but also the total moment of inertia tensor (Baerlocher and Boulic, 2000) or the total angular momentum (Kajita et al., 2003) in order to generate whole-body motion. They are closely related to the equations of motion of the base-link concerning angular movement.

As can be seen, some humanoid studies utilize the equations of motion of the base-link as a low-dimensional model for whole-body control. For identification of legged systems, the equations of motion of the base-link are also expected to extract the mechanical properties. A few identification works for industrial manipulators use a force/moment sensor mounted on a fixed base (Liu et al., 1998). They make use of the mechanical structure of a fixed base and identify the parameters without the equations of motion corresponding to each joint coordinate. Although they also show the usefulness of the base-link in identification, the identifiability of whole base parameters in the case when using the floating-base dynamics has not yet been discussed.

1.4. Overview of this paper

This paper consists of two parts. In the first part (Sections 2 to 5) we discuss the mechanical properties of the equations of motion of the base-link of legged systems for dynamics identification.

The equations of motion of the base-link have the following mechanical characteristic. The minimal set of inertial parameters (or the base parameters (Mayeda et al., 1990; Gautier, 1990; Kawasaki et al., 1991; Khalil and Bennis, 1995)) used to describe the six equations of motion corresponding to the base-link of general legged systems is equivalent to that of the whole-body equations. This implies that the base-link equations of legged systems have necessary and sufficient information about inertial parameters to describe the whole-body motion, which provides the validity to control or analyze a legged system by utilizing the base-link dynamics as the low-dimensional model of the whole system. From the characteristic of the base-link, all of the base parameters of the whole-body dynamics can be identified only from the six base-link equations; we can identify the parameters without the joint torques and the nonlinear internal components such as joint friction and viscosity. We only require the measurement of the generalized external forces of the base-link and the whole generalized coordinates. Section 2 introduces the linear form of the equations of motion of multibody systems with respect to inertial parameters, and the general method to compose the base parameters. Section 3 gives the mathematical notation and definitions used for the proof of the identifiability in Section 4.

When the robot has no external reaction force through the interaction with the environment, for example while jumping or flying, the following dynamics constraint exists; the generalized input and external forces of the base-link of the robot are always equal to zero. As this constraint is related to the conservation of angular momentum which means a non-holonomic constraint, legged systems enter the category of underactuated mechanisms. The control and motion planning of such systems have been an active area of research in robotics (Dubowsky and Papadopoulos, 1993; Nakamura and Mukherjee, 1993). The non-holonomic property decreases the number of base parameters of the base-link equations; the total mass of the system is always unidentifiable from the base-link dynamics under the law of conservation of momentum. In the case of general legged systems, the decrease of the number of base parameters is only one, and all of the base parameters normalized by the total mass can be identified only from the measurement of the motion. In Section 5 we discuss the unidentifiable parameters of the legged systems under the law of conservation of momentum.

The latter part of this paper (Sections 6–8) presents the identification method of inertial parameters of legged systems by utilizing the mechanical feature of the base-link dynamics, and the experimental results.

In Section 6, the identification method without the measurement of joint torques or the estimation from actuator models is proposed for legged systems. Consequently, only measurement of the contact forces, the joint angles, and the generalized coordinates of the base-link are required; the information can be measured not only by the internal sensors such as force/moment sensors, encoders, gyroscopes, and acceleration sensors, but also by the external sensors located in the environment such as force plates and a motion capture system. As only the six equations of the base-link are used, the computational cost is quite low.

Section 6 also presents the identification method without the measurement of external force. The method requires no force sensors but using instead the motion data under the law of conservation of momentum. It means that the identified parameters are unaffected by the measurement error or noise of forces.

Section 7 shows the experimental identification results of a small-size humanoid robot, and Section 8 gives those of two floating systems: a simple chain consists of two links and the jumping human body dynamics.

2. Linear form in the parameters of the dynamics for legged mechanisms

The equations of motion of legged systems, composed of N_L rigid bodies and that has N_J generalized coordinates, or $N_J - 6$ DOFs, are given by Yoshida et al. (1995); Fujimoto et al. (1998)

$$\begin{bmatrix} \boldsymbol{H}_{O1} \, \boldsymbol{H}_{O2} \\ \boldsymbol{H}_{C1} \, \boldsymbol{H}_{C2} \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{q}}_0 \\ \ddot{\boldsymbol{q}}_c \end{bmatrix} + \begin{bmatrix} \boldsymbol{b}_O \\ \boldsymbol{b}_C \end{bmatrix} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{\tau} \end{bmatrix} + \sum_{k=1}^{N_c} \begin{bmatrix} \boldsymbol{J}_{Ok}{}^{\mathrm{T}} \\ \boldsymbol{J}_{Ck}{}^{\mathrm{T}} \end{bmatrix} \boldsymbol{F}_k^{ext} \tag{1}$$

where:

- $\boldsymbol{H}_{ij}$ $(i = O, C,\ j = 1, 2)$ is the inertia matrix;

- $\boldsymbol{q}_0 \in SE(3)$ is the vector of the generalized coordinates which represent the position and orientation of the base-link;
- $\boldsymbol{q}_c \in \mathbb{R}^{N_J-6}$ is the joint angle vector;
- $\boldsymbol{b}_i$ is the bias force vector including centrifugal, Coriolis, and gravity forces;
- $\boldsymbol{\tau} \in \mathbb{R}^{N_J-6}$ is the vector of joint torques;
- N_c is the number of contact points with the environment;
- $\boldsymbol{F}_k^{ext} \in \mathbb{R}^6$ is the vector of external forces exerted on the system at contact k;
- $\boldsymbol{J}_k \triangleq [\boldsymbol{J}_{Ok} \ \ \boldsymbol{J}_{Ck}] \in \mathbb{R}^{6\times N_J}$ is the basic Jacobian matrix (Khatib, 1987) associated with contact k and of the orientation of the contact link with respect to generalized coordinates.

The upper part of Equation (1) represents the dynamics of the position and orientation of the base-link, which means the root of the kinematic tree structure. The lower part corresponds to the joint coordinates. The equations of motion of multibody systems can be written in a linear form with respect to the inertial parameters (Mayeda et al. 1984; Atkeson et al. 1986), and the following equation can be obtained from Equation (1):

$$\begin{bmatrix} \boldsymbol{Y}_O \\ \boldsymbol{Y}_C \end{bmatrix} \boldsymbol{\phi} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{\tau} \end{bmatrix} + \sum_{k=1}^{N_c} \begin{bmatrix} \boldsymbol{J}_{Ok}{}^{\mathrm{T}} \\ \boldsymbol{J}_{Ck}{}^{\mathrm{T}} \end{bmatrix} \boldsymbol{F}_k^{ext} \tag{2}$$

where:

- $\boldsymbol{\phi} \in \mathbb{R}^{10N_L}$ is the vector of constant inertial parameters such that

$$\boldsymbol{\phi} = \left[{}^0\boldsymbol{\phi}_0{}^{\mathrm{T}} \ {}^1\boldsymbol{\phi}_1{}^{\mathrm{T}} \ .. \ {}^{N_L-1}\boldsymbol{\phi}_{N_L-1}{}^{\mathrm{T}} \right]^{\mathrm{T}} \tag{3}$$

- ${}^i\boldsymbol{\phi}_i \in \mathbb{R}^{10}$ is the vector of constant inertial parameters of link i expressed in the coordinates system attached to link i (the details of ${}^i\boldsymbol{\phi}_i$ is shown in the next section);
- $\boldsymbol{Y}_A \triangleq [\boldsymbol{Y}_O{}^{\mathrm{T}} \ \ \boldsymbol{Y}_C{}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{N_J\times 10N_L}$ is the observation matrix or regressor, which is a function of $\boldsymbol{q}_0$, $\boldsymbol{q}_c$, their derivatives, and geometric parameters such as the length of each link; the method to obtain $\boldsymbol{Y}_A$ is shown in Atkeson et al. (1986).

Only the minimal set of inertial parameters can be identified from the linear form. This minimal set is called the base parameters. It is computed symbolically or numerically from the inertial parameters $\boldsymbol{\phi}$ by eliminating those that have no influence on the model and regrouping some according to the kinematics of the system (Mayeda et al., 1990; Gautier, 1990; Kawasaki et al., 1991; Khalil and Bennis, 1995).

Let us detail the general computation method of the base parameters. Let $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}) \in \mathbb{R}^{M\times N}$ be the function matrix of independent variables $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$. (For example, in the case of (2), we can assume that $\boldsymbol{Y} = [\boldsymbol{Y}_O^{\mathrm{T}} \ \boldsymbol{Y}_C^{\mathrm{T}}]^{\mathrm{T}}$, $N = 10N_L$, and $M = N_J$.) Let be $\mathrm{V}(\boldsymbol{Y})$ be the vector-space spanned by the functional column vectors $\boldsymbol{y}_j$ $(1 \le j \le N)$ of the matrix $\boldsymbol{Y}$ as follows:

$$\mathrm{V}(\boldsymbol{Y}) \triangleq \mathrm{span}[\boldsymbol{y}_1, \boldsymbol{y}_2, \cdots, \boldsymbol{y}_N] \tag{4}$$

If $\forall \boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$, all vectors in the linear space $\mathrm{V}(\boldsymbol{Y})$ are presented as linear combinations of the linear independent basis $\boldsymbol{y}_{Bj} \in \mathbb{R}^M$ $(1 \le j \le N_B)$, then the dimension of the vector space $\mathrm{V}(\boldsymbol{Y})$ is $\dim \mathrm{V}(\boldsymbol{Y}) = N_B$. It should be noted that $\dim \mathrm{V}(\boldsymbol{Y})$ is different from $\mathrm{rank}(\boldsymbol{Y})$. It can be regarded as the maximum rank of the matrix which is the vertical concatenation of $\boldsymbol{Y}$ for the different sets of the values of $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$. Then the following proposition holds.

Proposition 1. *If* $\exists \boldsymbol{X} \in \mathbb{R}^{N\times N_B}$*, and if* $\boldsymbol{Y}_B \in \mathbb{R}^{M\times N_B}$ *defined by*

$$\boldsymbol{Y}_B \triangleq \boldsymbol{Y}\boldsymbol{X} \tag{5}$$

satisfies

$$\dim \mathrm{V}(\boldsymbol{Y}) = \dim \mathrm{V}(\boldsymbol{Y}_B) = N_B \tag{6}$$

then $\exists!\ \boldsymbol{Z} \in \mathbb{R}^{N_B\times N}$ *which verifies*

$$\boldsymbol{Y} = \boldsymbol{Y}_B\boldsymbol{Z} \tag{7}$$

and $\boldsymbol{\phi}_B \in \mathbb{R}^{N_B}$ *defined by*

$$\boldsymbol{\phi}_B \triangleq \boldsymbol{Z}\boldsymbol{\phi} \tag{8}$$

are the base parameters.

In order to compute the base parameters, we have to find the composition matrices $\boldsymbol{X}$ and $\boldsymbol{Z}$, which verify Equations (6) and (7).

The minimal parameter linear form is written as follows:

$$\begin{bmatrix} \boldsymbol{Y}_{OB} \\ \boldsymbol{Y}_{CB} \end{bmatrix} \boldsymbol{\phi}_B = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{\tau} \end{bmatrix} + \sum_{k=1}^{N_c} \begin{bmatrix} \boldsymbol{J}_{Ok}{}^{\mathrm{T}} \\ \boldsymbol{J}_{Ck}{}^{\mathrm{T}} \end{bmatrix} \boldsymbol{F}_k^{ext} \tag{9}$$

where:

- N_B is the number of the base parameters;
- $\boldsymbol{\phi}_B \in \mathbb{R}^{N_B}$ is the vector of the base parameters;
- $\boldsymbol{Y}_{OB} \in \mathbb{R}^{6\times N_B}$, $\boldsymbol{Y}_{CB} \in \mathbb{R}^{(N_J-6)\times N_B}$ are the regressor matrices for the base parameters.

Here, let us extract only the upper part of Equation (9), the equations of motion of the base-link:

$$\boldsymbol{Y}_{OB}\boldsymbol{\phi}_B = \sum_{k=1}^{N_c} \boldsymbol{J}_{Ok}{}^{\mathrm{T}} \boldsymbol{F}_k^{ext} \tag{10}$$

The linear form given by Equation (10) reduces the whole system to these six equations. The reduction of the system to the six equations of the base-link keeps the number of structurally identifiable parameters unchanged. In the following sections, we prove this important feature of the equations of motion of the base-link.

3. Preliminary notation

3.1. Notations of open kinematic chains

The notation to be used are shown in Figure 1 for an open tree-structure.

- The indices are chosen from the base link to the end of branches, so that index j is given to a link which is attached to link L_i where $0 \leq i < j$, i.e. a link already indexed.
- Link L_0 is the base-link.
- C_i is the sub-chain composed of links L_0 to L_i.
- p_i is the index of the parent link of L_i in C_i.
- The set of first rank children of L_i is $\mathcal{J}_{i,i}$, while the whole set of children of L_i is $\mathcal{J}_i$.
- $n_{J,i}$ is the number of DOFs of joint i. $N_{J,i} = \sum_k^{k \in C_i} n_{J,k}$ is the number of DOFs of sub-chain C_i, and when $i = n-1$ we note $N_{J,N} = N_J$.
- $\Sigma_w (= \Sigma_{p0})$ is the world absolute coordinates system.
- Σ_i is the coordinates system attached to L_i.
- $\boldsymbol{q}_i \in \mathbb{R}^{n_{J,i}}$ is the vector of generalized coordinates corresponding to joint i.
- $\boldsymbol{\tau}_i \in \mathbb{R}^{n_{J,i}}$ is the vector of generalized forces associated with joint i.
- ${}^i\boldsymbol{R}_j \in SO(3)$ is the rotation matrix from Σ_i to Σ_j.
- ${}^i\boldsymbol{p}_j^i \in \mathbb{R}^3$ is the translation vector from Σ_i to Σ_j given in Σ_i.
- ${}^i\dot{\boldsymbol{p}}_i, {}^i\ddot{\boldsymbol{p}}_i \in \mathbb{R}^3$ are the vectors of translational velocity and acceleration of L_i in Σ_i.
- ${}^i\boldsymbol{\omega}_i, {}^i\dot{\boldsymbol{\omega}}_i \in \mathbb{R}^3$ are the vectors of angular velocity and acceleration of L_i in Σ_i .
- $m_i \in \mathbb{R}$ is the mass of L_i.
- ${}^i\boldsymbol{s}_i \in \mathbb{R}^3$ is the coordinate of the COM of L_i in Σ_i.
- ${}^i\boldsymbol{I}_i \in \mathbb{R}^{3\times 3}$ is the inertia matrix of L_i around Σ_i.
- $\boldsymbol{K}_i \in \mathbb{R}^{n_{J,i}\times 6}$ is the matrix of orthogonal projection of forces ${}^i\boldsymbol{f}_i \in \mathbb{R}^6$ working on joint i and such that $\tau_i = \boldsymbol{K}_i {}^i\boldsymbol{f}_i$.
- ${}^i\boldsymbol{\phi}_i \in \mathbb{R}^{10}$ is the vector of inertial parameters of L_i expressed in Σ_i, such as

$$
\begin{aligned}
{}^i\boldsymbol{\phi}_i \triangleq [\, m_i \;\; m_i{}^is_{i,x} \;\; m_i{}^is_{i,y} \;\; m_i{}^is_{i,z} \;\; {}^iI_{i,xx} \\
{}^iI_{i,yy} \;\; {}^iI_{i,zz} \;\; {}^iI_{i,yz} \;\; {}^iI_{i,zx} \;\; {}^iI_{i,xy} \,]^{\mathrm{T}}
\end{aligned} \tag{11}
$$

- $[\boldsymbol{\omega}\times]$ and $[\boldsymbol{\omega}\bullet]$ are given by

$$
[\boldsymbol{\omega}\times] \triangleq \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \tag{12}
$$

$$
[\boldsymbol{\omega}\bullet] \triangleq \begin{bmatrix} \omega_x & 0 & 0 & 0 & \omega_z & \omega_y \\ 0 & \omega_y & 0 & \omega_z & 0 & \omega_x \\ 0 & 0 & \omega_z & \omega_y & \omega_x & 0 \end{bmatrix} \tag{13}
$$

- $\mathrm{diag}(\boldsymbol{x})$ denotes the diagonal matrix with vector $\boldsymbol{x}$ on the diagonal.
- $\boldsymbol{0}$ is the zero vector, $\boldsymbol{O}$ denotes the zero matrix, and $\boldsymbol{E}$ represents the identity matrix.

We assume that $\boldsymbol{K}_i$ is constant, and each row vector of $\boldsymbol{K}_i$ is orthogonal to all others. By using $\boldsymbol{K}_i$, the relative velocity of joint i is expressed as

$$
\begin{bmatrix} {}^i\dot{\boldsymbol{p}}_i^{p_i} \\ {}^i\boldsymbol{\omega}_i^{p_i} \end{bmatrix} = \boldsymbol{K}_i^{\mathrm{T}}\dot{\boldsymbol{q}}_i = \sum_{k=1}^{n_{J,i}} \begin{bmatrix} \boldsymbol{e}_{pi,k} \\ \boldsymbol{e}_{qi,k} \end{bmatrix} \dot{q}_{i,k} \tag{14}
$$

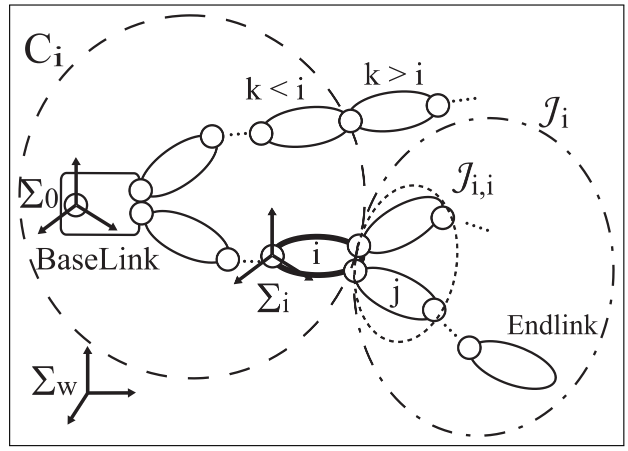


Fig. 1. Notation of the open kinematic chain.

where ${}^i\dot{\boldsymbol{p}}_i^{p_i} \in \mathbb{R}^3$ is the vector of relative linear velocity of joint i, ${}^i\boldsymbol{\omega}_i^{p_i} \in \mathbb{R}^3$ is the vector of relative angular velocity of joint i, $\dot{q}_{i,k}$ is the kth element of $\dot{\boldsymbol{q}}_i$ ($1 \leq k \leq n_{J,i}$), $\boldsymbol{k}_{i,k} \in \mathbb{R}^6$ is the kth row of $\boldsymbol{K}_i$, and we define $\boldsymbol{e}_{pi,k} \in \mathbb{R}^3$ and $\boldsymbol{e}_{qi,k} \in \mathbb{R}^3$ as follows:

$$
\begin{bmatrix} \boldsymbol{e}_{pi,k} \\ \boldsymbol{e}_{qi,k} \end{bmatrix} \triangleq \boldsymbol{k}_{i,k}{}^{\mathrm{T}} \tag{15}
$$

3.2. Regressor matrix for the sub-chain C_j

We now consider the sub-chain C_j, and its inertial parameters $\boldsymbol{\phi}_j \in \mathbb{R}^{10N_j}$ is given by

$$
\boldsymbol{\phi}_j \triangleq \left[{}^0\boldsymbol{\phi}_0{}^{\mathrm{T}} \; {}^1\boldsymbol{\phi}_1{}^{\mathrm{T}} \cdots {}^j\boldsymbol{\phi}_j{}^{\mathrm{T}} \right]^{\mathrm{T}} \tag{16}
$$

The corresponding regressor for the sub-chain C_j are $\boldsymbol{Y}_{Oj}$ and $\boldsymbol{Y}_{Aj}$, where $\boldsymbol{Y}_{Oj} \in \mathbb{R}^{6\times 10(j+1)}$ is obtained using only the six equations of the base-link, and $\boldsymbol{Y}_{Aj} \in \mathbb{R}^{N_{J,j}\times 10(j+1)}$ is obtained using the $N_{J,j}$ equations of the whole sub-chain C_j. They are defined respectively as follows:

$$
\boldsymbol{Y}_{Oj} \triangleq \begin{cases} {}^0\boldsymbol{Y}_0 & (j=0) \\ \left[\boldsymbol{Y}_{Oj-1} \;\; {}^0\boldsymbol{A}_j{}^j\boldsymbol{Y}_j \right] & (j \neq 0) \end{cases} \tag{17}
$$

$$
\boldsymbol{Y}_{Aj} \triangleq \begin{cases} {}^0\boldsymbol{Y}_0 & (j=0) \\ \begin{bmatrix} \boldsymbol{Y}_{Aj-1} & {}^0\widetilde{\boldsymbol{A}}_{Aj}{}^j\boldsymbol{Y}_j \\ \boldsymbol{O} & \boldsymbol{K}_j{}^j\boldsymbol{Y}_j \end{bmatrix} & (j \neq 0) \end{cases} \tag{18}
$$

where:

- ${}^j\boldsymbol{Y}_j \in \mathbb{R}^{6\times 10}$ is the individual regressor matrix of the link j (Atkeson et al., 1986) such that

$$
{}^j\boldsymbol{Y}_j \triangleq \begin{bmatrix} {}^j\ddot{\boldsymbol{p}}_j & [{}^j\dot{\boldsymbol{\omega}}_j\times] + [{}^j\boldsymbol{\omega}_j\times]^2 & \boldsymbol{O} \\ \boldsymbol{0} & -[{}^j\ddot{\boldsymbol{p}}_j\times] & [{}^j\dot{\boldsymbol{\omega}}_j\bullet] + [{}^j\boldsymbol{\omega}_j\times][{}^j\boldsymbol{\omega}_j\bullet] \end{bmatrix} \tag{19}
$$

- ${}^i\boldsymbol{A}_j \in \mathbb{R}^{6\times 6}$ and ${}^i\widetilde{\boldsymbol{A}}_j \in \mathbb{R}^{6\times 6}$ are the matrices corresponding in the coordinates change for the forces from Σ_j to Σ_i, such that

$$ {}^i\boldsymbol{A}_j \triangleq \begin{bmatrix} {}^i\boldsymbol{R}_j & \boldsymbol{O} \\ [{}^i\boldsymbol{p}^i_j\times]\,{}^i\boldsymbol{R}_j & {}^i\boldsymbol{R}_j \end{bmatrix} \tag{20} $$

$$ {}^i\widetilde{\boldsymbol{A}}_j \triangleq \begin{cases} \boldsymbol{E} & (j = i) \\ {}^i\boldsymbol{A}_j & (j \in \mathcal{J}_i, j \neq i) \\ \boldsymbol{O} & (j \notin \mathcal{J}_i, j \neq i) \end{cases} \tag{21} $$

- ${}^0\widetilde{\boldsymbol{A}}_{A_j} \in \mathbb{R}^{N_{J,j-1}\times 6}$ is given by

$$ {}^0\widetilde{\boldsymbol{A}}_{A_j} \triangleq \left[{}^0\boldsymbol{A}_j^{\mathrm{T}}\,(\boldsymbol{K}_1{}^1\widetilde{\boldsymbol{A}}_j)^{\mathrm{T}} \cdots (\boldsymbol{K}_{j-1}{}^{j-1}\widetilde{\boldsymbol{A}}_j)^{\mathrm{T}} \right]^{\mathrm{T}} \tag{22} $$

3.3. *Coordinate transformations for the inertial parameters*

The inertial parameters ${}^j\boldsymbol{\phi}_j$ of L_j can be transformed into ${}^i\boldsymbol{\phi}_j$ expressed in Σ_i as follows:

$$ {}^i\boldsymbol{\phi}_j = {}^i\boldsymbol{B}_j{}^j\boldsymbol{\phi}_j \tag{23} $$

${}^i\boldsymbol{B}_j \in \mathbb{R}^{10\times 10}$ is the transformation matrix of the inertial parameters (Khalil and Bennis, 1995). Let p_i and $r_{ij}(1 \leq i, j \leq 3)$ be each element of $\boldsymbol{p}$ and $\boldsymbol{R}$. Here ${}^i\boldsymbol{B}_j$ is defined as follows:

$$ {}^i\boldsymbol{B}_j \triangleq \begin{bmatrix} 1 & \boldsymbol{0}^{\mathrm{T}} & \boldsymbol{0}^{\mathrm{T}} \\ {}^i\boldsymbol{p}^i_j & {}^i\boldsymbol{R}_j & \boldsymbol{O} \\ \boldsymbol{B}_M({}^i\boldsymbol{p}^i_j) & \boldsymbol{B}_C({}^i\boldsymbol{p}^i_j){}^i\boldsymbol{R}_j & \boldsymbol{B}_I({}^i\boldsymbol{R}_j) \end{bmatrix} \tag{24} $$

with

$$ \boldsymbol{B}_M(\boldsymbol{p}) \triangleq \begin{bmatrix} p_2{}^2 + p_3{}^2 & p_3{}^2 + p_1{}^2 & p_1{}^2 + p_2{}^2 & -p_2p_3 & -p_3p_1 & -p_1p_2 \end{bmatrix}^{\mathrm{T}} \tag{25} $$

$$ \boldsymbol{B}_C(\boldsymbol{p}) \triangleq \begin{bmatrix} 0 & 2p_1 & 2p_1 & 0 & -p_3 & -p_2 \\ 2p_2 & 0 & 2p_2 & -p_3 & 0 & -p_1 \\ 2p_3 & 2p_3 & 0 & -p_2 & -p_1 & 0 \end{bmatrix}^{\mathrm{T}} \tag{26} $$

$$ \boldsymbol{B}_I(\boldsymbol{R}) \triangleq \begin{bmatrix} r_{11}{}^2 & r_{12}{}^2 & r_{13}{}^2 & 2r_{12}r_{13} & 2r_{13}r_{11} & 2r_{11}r_{12} \\ r_{21}{}^2 & r_{22}{}^2 & r_{23}{}^2 & 2r_{22}r_{23} & 2r_{23}r_{21} & 2r_{21}r_{22} \\ r_{31}{}^2 & r_{32}{}^2 & r_{33}{}^2 & 2r_{32}r_{33} & 2r_{33}r_{31} & 2r_{31}r_{32} \\ r_{21}r_{31} & r_{22}r_{32} & r_{23}r_{33} & r_{22}r_{33} + r_{32}r_{23} & r_{23}r_{31} + r_{33}r_{21} & r_{21}r_{32} + r_{31}r_{22} \\ r_{31}r_{11} & r_{32}r_{12} & r_{33}r_{13} & r_{32}r_{13} + r_{12}r_{33} & r_{33}r_{11} + r_{13}r_{31} & r_{31}r_{12} + r_{11}r_{32} \\ r_{11}r_{21} & r_{12}r_{22} & r_{13}r_{23} & r_{12}r_{23} + r_{22}r_{13} & r_{13}r_{21} + r_{23}r_{11} & r_{11}r_{22} + r_{21}r_{12} \end{bmatrix} \tag{27} $$

By using ${}^i\boldsymbol{B}_j$, the individual regressor ${}^j\boldsymbol{Y}_j$ is decomposed as follows:

$$ {}^j\boldsymbol{Y}_j = {}^{p_j}\boldsymbol{A}_j^{-1}\,{}^{p_j}\boldsymbol{Y}_{p_j}\,{}^{p_j}\boldsymbol{B}_j + {}^j\boldsymbol{Y}^{p_j}_j \tag{28} $$

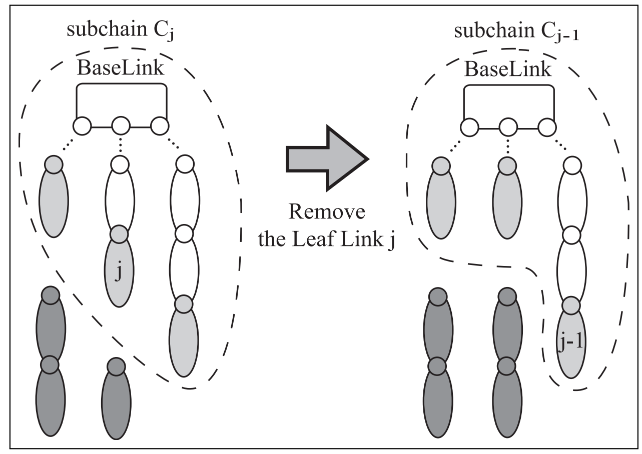


Fig. 2. The outline of ordinary calculation method of the base parameters from the end-link side. The base parameters of the gray links have already been calculated.

where ${}^j\boldsymbol{Y}^{p_j}_j \in \mathbb{R}^{6\times 10}$ is given by

$$ \begin{aligned} {}^j\boldsymbol{Y}^{p_j}_j \triangleq\ & \boldsymbol{Y}_m({}^j\dot{\boldsymbol{p}}_j - {}^j\dot{\boldsymbol{p}}^{p_j}_j, {}^j\boldsymbol{\omega}_j - {}^j\boldsymbol{\omega}^{p_j}_j)\,{}^j\boldsymbol{B}^{p_j}_{dj} \\ & + {}^j\boldsymbol{A}_{dj}\boldsymbol{Y}_m({}^j\dot{\boldsymbol{p}}^{p_j}_j, {}^j\boldsymbol{\omega}^{p_j}_j) \\ & + \boldsymbol{Y}_m({}^j\ddot{\boldsymbol{p}}^{p_j}_j + {}^j\dot{\boldsymbol{p}}^{p_j}_j \times {}^j\boldsymbol{\omega}^{p_j}_j, {}^j\dot{\boldsymbol{\omega}}^{p_j}_j) \end{aligned} \tag{29} $$

with

$$ \boldsymbol{Y}_m(\boldsymbol{x}, \boldsymbol{y}) \triangleq \begin{bmatrix} \boldsymbol{x} & [\boldsymbol{y}\times] & \boldsymbol{O} \\ \boldsymbol{0} & -[\boldsymbol{x}\times] & [\boldsymbol{y}\bullet] \end{bmatrix} \tag{30} $$

$$ {}^j\boldsymbol{A}_{dj} \triangleq \begin{bmatrix} [{}^j\boldsymbol{\omega}_j\times] & \boldsymbol{O} \\ [{}^j\dot{\boldsymbol{p}}_j\times] & [{}^j\boldsymbol{\omega}_j\times] \end{bmatrix} \tag{31} $$

$$ {}^j\boldsymbol{B}^{p_j}_{dj} \triangleq \begin{bmatrix} 0 & \boldsymbol{0}^{\mathrm{T}} & \boldsymbol{0}^{\mathrm{T}} \\ {}^j\dot{\boldsymbol{p}}^{p_j}_j & [{}^j\boldsymbol{\omega}^{p_j}_j\times] & \boldsymbol{O} \\ \boldsymbol{0} & \boldsymbol{B}_C({}^j\dot{\boldsymbol{p}}^{p_j}_j) & \boldsymbol{B}_W({}^j\boldsymbol{\omega}^{p_j}_j) \end{bmatrix} \tag{32} $$

$$ \boldsymbol{B}_W(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & 0 & 0 & 0 & 2\omega_y & -2\omega_z \\ 0 & 0 & 0 & -2\omega_x & 0 & 2\omega_z \\ 0 & 0 & 0 & 2\omega_x & -2\omega_y & 0 \\ 0 & \omega_x & -\omega_x & 0 & \omega_z & -\omega_y \\ -\omega_y & 0 & \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & -\omega_z & 0 & \omega_y & -\omega_x & 0 \end{bmatrix} \tag{33} $$

4. The base parameters of the base-link dynamics

4.1. *Outline of the proposed method to compute the base parameters*

To compute the base parameters, the backward recursive method shown in Figure 2 is usually used in the presence of the torque measurement (Mayeda et al., 1990; Kawasaki et al., 1991). However, we propose to use a forward recursive method as shown in Figure 3, and thus prove recursively that $\boldsymbol{\phi}_B$ is identifiable by using Equation (10).

With the forward recursive method, first the base parameters of sub-chain C_{j-1} are computed, then the following

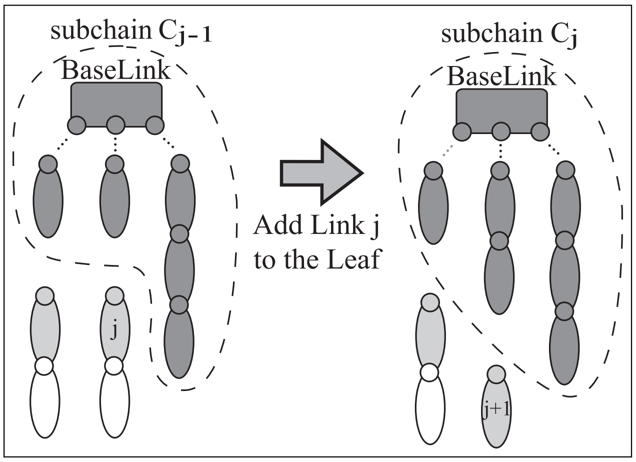


Fig. 3. The outline of proposed calculation method of the base parameters from base-link side.

link L_j is added, and the new base parameters in sub-chain C_j are computed. We first give the computation algorithm of the base parameters with the equations of motion of the base-link in Section 4.2. Then we give the algorithm of the base parameters with the whole system equations in Section 4.3. We prove that the obtained base parameters are the same.

4.2. Computation of the base parameters with the equations of dynamics of the base-link

4.2.1. Preliminaries We compute the base parameters of the whole system $\boldsymbol{\phi}_{OB} \in \mathbb{R}^{N_{OB}}$ from the six equations of the base-link with a forward recursive method. We compute recursively the composition matrix of the base parameters for sub-chain C_j. According to Proposition 1 for sub-chain C_j, we first choose a matrix $\boldsymbol{X}_{Oj} \in \mathbb{R}^{10(j+1)\times N_{OBj}}$ that satisfies Equation (34). Then only one matrix $\boldsymbol{Z}_{Oj} \in \mathbb{R}^{N_{OBj}\times 10(j+1)}$ satisfies Equation (35).

$$\dim \mathrm{V}(\boldsymbol{Y}_{Oj}) = \dim \mathrm{V}(\boldsymbol{Y}_{OBj}) = N_{OBj} \tag{34}$$

$$\boldsymbol{Y}_{Oj} = \boldsymbol{Y}_{OBj}\boldsymbol{Z}_{Oj} \tag{35}$$

By choosing $\boldsymbol{X}_{Oj}$ and $\boldsymbol{Z}_{Oj}$, the base parameters $\boldsymbol{\phi}_{OBj} \in \mathbb{R}^{N_{OBj}}$ and the corresponding regressor $\boldsymbol{Y}_{OBj} \in \mathbb{R}^{6\times N_{OBj}}$ are given by

$$\boldsymbol{Y}_{OBj} \triangleq \boldsymbol{Y}_{Oj}\boldsymbol{X}_{Oj} \tag{36}$$

$$\boldsymbol{\phi}_{OBj} \triangleq \boldsymbol{Z}_{Oj}\boldsymbol{\phi}_j \tag{37}$$

4.2.2. Initialization with C_0 We start the forward recursive computation with $j = 0$. Here C_0 is made of the base-link L_0 only. Now, let us define the following lemma. The proof of the lemma is given in Appendix A.

Lemma 1. *If a rigid body can move freely in the space, then,* $\dim \mathrm{V}(\boldsymbol{Y}_{O0}) = 10$ *holds.*

According to Lemma 1, the number of base parameters for the base-link is $N_{OB0} = 10$. Consequently we can choose $\boldsymbol{X}_{O0} = \boldsymbol{E}$ to satisfy $\boldsymbol{Y}_{O0} = \boldsymbol{Y}_{OB0}$ from Equation (34). Hence, Proposition Π_{O0} holds.

Proposition Π_{O0}. *For sub-chain C_0 made of L_0, $\boldsymbol{X}_{O0}$, and $\boldsymbol{Z}_{O0}$, defined by*

$$\boldsymbol{X}_{O0} = \boldsymbol{Z}_{O0} \triangleq \boldsymbol{E} \tag{38}$$

verifies Equations (34) and (35) ($N_{OB0} = n_{B0} = 10$).

4.2.3. General case for sub-chain C_j Let us now consider sub-chain C_j for $j > 0$. To compute the base parameters, we present Proposition Π_{Oj}.

Proposition Π_{Oj}. *If Proposition Π_{Oj-1} holds, and if $\boldsymbol{X}_{j,j} \in \mathbb{R}^{10\times n_{Bj}}$ and $\boldsymbol{X}_{j,j}^{\perp} \in \mathbb{R}^{10\times(10-n_{Bj})}$ satisfy Condition Ψ_j, then the constant matrices $\boldsymbol{X}_{Oj}$ and $\boldsymbol{Z}_{Oj}$, defined by*

$$\boldsymbol{X}_{Oj} \triangleq \begin{bmatrix} \boldsymbol{X}_{Oj-1} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{X}_{j,j} \end{bmatrix} \tag{39}$$

$$\boldsymbol{Z}_{Oj} \triangleq \begin{bmatrix} \boldsymbol{Z}_{Oj-1} & \boldsymbol{Z}_{Oj-1,i}{}^{p_j}\boldsymbol{B}_j\boldsymbol{X}_{j,j}^{\perp}\boldsymbol{X}_{j,j}^{\perp\mathrm{T}} \\ \boldsymbol{O} & \boldsymbol{X}_{j,j}{}^{\mathrm{T}} \end{bmatrix} \tag{40}$$

verify Equation (34) and (35) ($N_{OBj} = N_{OBj-1} + n_{Bj}$).

Condition Ψ_j is defined as follows.

Condition $\Psi_j (j > 0)$. All of the columns $\boldsymbol{x}_{j,m}$ ($1 \le m \le n_{Bj}$) of $\boldsymbol{X}_{j,j} \in \mathbb{R}^{10\times n_{Bj}}$ are an orthogonal basis of the vector-space V_j which is the orthogonal complementary space of $V_j^{\perp}$ spanned by the vectors $\boldsymbol{x}_j^{\perp}$ which satisfy

$$\boldsymbol{W}_{j,k}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \tag{41}$$

$$\boldsymbol{W}_{j,k,l}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \tag{42}$$

($\forall k,\ l \in [1\ n_j]$) where $\boldsymbol{W}_{j,k} \in \mathbb{R}^{12\times 10}$ and $\boldsymbol{W}_{j,k,l} \in \mathbb{R}^{6\times 10}$ are the constant matrices defined as follows:

$$\boldsymbol{W}_{j,k} \triangleq \begin{bmatrix} \boldsymbol{e}_{pj,k} & [\boldsymbol{e}_{qj,k}\times] & \boldsymbol{O} \\ \boldsymbol{0} & -[\boldsymbol{e}_{pj,k}\times] & [\boldsymbol{e}_{qj,k}\bullet] \\ \boldsymbol{0} & \boldsymbol{B}_C(\boldsymbol{e}_{pj,k}) & \boldsymbol{B}_W(\boldsymbol{e}_{qj,k}) \end{bmatrix} \tag{43}$$

$$\boldsymbol{W}_{j,k,l} \triangleq \begin{bmatrix} \boldsymbol{e}_{pj,k}\times\boldsymbol{e}_{qj,k} & \boldsymbol{O} & \boldsymbol{O} \\ \boldsymbol{0} & -[(\boldsymbol{e}_{pj,k}\times\boldsymbol{e}_{qj,k})\times] & \boldsymbol{O} \end{bmatrix} \tag{44}$$

Let us define the following vectors and matrix for the proof of Proposition Π_{Oj}:

- $\boldsymbol{a}_j \in \mathbb{R}^{N_{OBj}}$ is an arbitrary constant vector corresponding to sub-chain C_j;
- $\boldsymbol{a}_{j,j} \in \mathbb{R}^{n_{Bj}}$ is a sub-vector of $\boldsymbol{a}_j$, and is corresponding to link j; $\boldsymbol{a}_j$ can be written recursively as follows:

$$\boldsymbol{a}_j \triangleq [\boldsymbol{a}_{j-1}^{\mathrm{T}}\ \ \boldsymbol{a}_{j,j}^{\mathrm{T}}]^{\mathrm{T}} \tag{45}$$

- $\boldsymbol{x}_j \in \mathbb{R}^{10}$ is defined by

$$\boldsymbol{x}_j \triangleq \boldsymbol{X}_{j,j}\boldsymbol{a}_{j,j} \tag{46}$$

- $\boldsymbol{Z}_{Oj,k} \in \mathbb{R}^{N_{OBj-1}\times 10}$ is made of the 10 columns of $\boldsymbol{Z}_{Oj}$ corresponding to link k ($0 \leq k \leq j$); by using the columns $z_{Oj,l} \in \mathbb{R}^{N_{OBj}}(1 \leq l \leq 10(j+1))$, $\boldsymbol{Z}_{Oj,k}$ is given by

$$\boldsymbol{Z}_{Oj,k} \triangleq \left[z_{Oj,10k+1} \cdots z_{Oj,10k+10} \right] \tag{47}$$

Let us define the following lemmas that are used for the proof of Proposition Π_{Oj}. The proofs of those lemmas are shown in Appendices B and C.

Lemma 2. *The necessary and sufficient condition that* $\boldsymbol{x}_j^{\perp}$ *satisfies*

$${}^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \tag{48}$$

for all possible $\boldsymbol{q}_0$, $\boldsymbol{q}_c$, *and their derivatives, is that* $\boldsymbol{x}_j^{\perp}$ *satisfies both Equations (41) and (42).*

Lemma 3. *If* $\boldsymbol{x}_j^{\perp}$ *satisfies Equations (41) and (42), then*

$${}^{p_j}\boldsymbol{B}_j\boldsymbol{x}_j^{\perp} = const \tag{49}$$

holds.

Proof of Π_{Oj}. We first verify the condition when $\boldsymbol{X}_{Oj}$ does not satisfy Equation (34):

$$\dim(\boldsymbol{Y}_{Oj}\boldsymbol{X}_{Oj}) \neq N_{OBj} \tag{50}$$

When $\exists \boldsymbol{a}_j \neq \boldsymbol{0}, \forall \boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$,

$$\boldsymbol{Y}_{Oj}\boldsymbol{X}_{Oj}\boldsymbol{a}_j = \boldsymbol{0} \tag{51}$$

holds from Equation (50). By using Equations (17), (28), (39), (40), (47), and (35) of sub-chain C_{j-1}, Equation (51) is modified as

$$\boldsymbol{Y}_{OBj-1}(\boldsymbol{a}_{j-1} + \boldsymbol{Z}_{Oj-1,p_j}{}^{p_j}\boldsymbol{B}_j\boldsymbol{x}_j) + {}^{0}\boldsymbol{A}_j{}^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{x}_j = \boldsymbol{0} \tag{52}$$

Since Equation (52) holds for all $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$, it is also verified when $\dot{\boldsymbol{q}}_j = \ddot{\boldsymbol{q}}_j = \boldsymbol{0}$ (i.e. ${}^{j}\boldsymbol{Y}_j^{p_j} = \boldsymbol{O}$). All terms except ${}^{j}\boldsymbol{Y}_j^{p_j}$ in Equation (52) are independent from $\dot{\boldsymbol{q}}_j$ and $\ddot{\boldsymbol{q}}_j$, and ${}^{0}\boldsymbol{A}_j$ is a regular matrix (rank(${}^{0}\boldsymbol{A}_j$)$= 6$). In addition, Equation (34) holds for sub-chain C_{j-1}, and ${}^{p_j}\boldsymbol{B}_j$ is independent from the coordinates of C_{j-1} Thus, Equation (52) is equivalent to both

$$\boldsymbol{Z}_{Oj-1,p_j}{}^{p_j}\boldsymbol{B}_j\boldsymbol{x}_j = const(= -\boldsymbol{a}_{j-1}) \tag{53}$$

$${}^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{x}_j = \boldsymbol{0} \tag{54}$$

According to Lemma 2, a necessary and sufficient condition to have Equation (54) is that

$$\boldsymbol{W}_{j,k}\boldsymbol{x}_{j,m} = \boldsymbol{0} \tag{55}$$

$$\boldsymbol{W}_{j,k,l}\boldsymbol{x}_{j,m} = \boldsymbol{0} \tag{56}$$

hold $\forall k,\ l \in [1\ n_j]$.

According to Lemma 3, if both Equations (55) and (56) hold, then Equations (53) is verified. Therefore, a necessary and sufficient condition to have Equation (50) is that Equations (55) and (56) hold $\forall k,\ l \in [1\ n_j]$.

From the converse of Equations (50), (55), and (56), if $\boldsymbol{X}_{j,j}$ satisfies Condition Ψ_j, then $\boldsymbol{X}_{Oj}$ defined by Equation (39) satisfies Equation (34).

Next, we verify that $\boldsymbol{Z}_{Oj}$ defined by Equation (40) is constant. The column vectors $\boldsymbol{x}_j^{\perp}$ of $\boldsymbol{X}_{j,j}^{\perp}$ satisfied Equations (41) and (42). Similarly to the case of Equations (53), the following relationship holds according to Lemma 3:

$$\boldsymbol{Z}_{Oj-1,p_j}{}^{p_j}\boldsymbol{B}_j\boldsymbol{X}_{j,j}^{\perp} = constant \tag{57}$$

Therefore, $\boldsymbol{Z}_{Oj}$ defined by Equation (40) is constant.

Finally, we show that $\boldsymbol{Z}_{Oj}$ defined by Equation (40) satisfies Equation (35).

$$\boldsymbol{Y}_{OBj}\boldsymbol{Z}_{Oj} = \left[\boldsymbol{Y}_{OBj-1}\boldsymbol{X}_{Oj-1} \quad \boldsymbol{V}_1 + \boldsymbol{V}_2 \right] \tag{58}$$

$$\begin{aligned} \boldsymbol{V}_1 &\triangleq \boldsymbol{Y}_{Oj-1}\boldsymbol{X}_{Oj-1}\boldsymbol{Z}_{Oj-1,p_j}{}^{p_j}\boldsymbol{B}_j\boldsymbol{X}_{j,j}^{\perp}\boldsymbol{X}_{j,j}^{\perp\mathrm{T}} \\ &= {}^{0}\boldsymbol{A}_{p_j}{}^{p_j}\boldsymbol{Y}_{p_j}{}^{p_j}\boldsymbol{B}_j\boldsymbol{X}_{j,j}^{\perp}\boldsymbol{X}_{j,j}^{\perp\mathrm{T}} \end{aligned} \tag{59}$$

$$\boldsymbol{V}_2 \triangleq {}^{0}\boldsymbol{A}_j{}^{j}\boldsymbol{Y}_j\boldsymbol{X}_{j,j}\boldsymbol{X}_{j,j}^{\mathrm{T}} \tag{60}$$

The columns of $\boldsymbol{X}_{j,j}$ and $\boldsymbol{X}_{j,j}^{\perp}$ are an orthogonal basis of 10-dimensional space such that

$$\boldsymbol{X}_{j,j}\boldsymbol{X}_{j,j}^{\mathrm{T}} + \boldsymbol{X}_{j,j}^{\perp}\boldsymbol{X}_{j,j}^{\perp\mathrm{T}} = \boldsymbol{E} \tag{61}$$

From Equation (54) and Condition Ψ_j, we obtain

$${}^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{X}_{j,j}^{\perp} = \boldsymbol{O} \tag{62}$$

From Equations (28), (36), (61), and (62), we obtain

$$\boldsymbol{V}_1 + \boldsymbol{V}_2 = {}^{0}\boldsymbol{A}_j{}^{j}\boldsymbol{Y}_j \tag{63}$$

From Equations (58), (63), and (35) for sub-chain C_{j-1}, then $\boldsymbol{Y}_{OBj}\boldsymbol{Z}_{Oj}$ is modified as

$$\boldsymbol{Y}_{OBj}\boldsymbol{Z}_{Oj} = \left[\boldsymbol{Y}_{Oj-1}\ {}^{0}\boldsymbol{A}_j{}^{j}\boldsymbol{Y}_j \right] \tag{64}$$

Since the right-hand side of Equation (64) is equivalent to $\boldsymbol{Y}_{Oj}$ from Equation (17), then Equation (35) holds.

From the above, Proposition Π_{Oj} is verified.

4.2.4. Summary From Proposition Π_{O0}, Π_{Oj}, the following proposition holds.

Proposition 2. *For all* $j \in [1\ n-1]$, *if* $\boldsymbol{X}_{j,j}$ *satisfies Condition* Ψ_j, $\boldsymbol{X}_{Oj}$ *and* $\boldsymbol{Z}_{Oj}$, *defined by Equations (38), (39), and (40), are the composition matrices of the base parameters of the six equations of motion of the base-link.*

4.3. Computation of the base parameters with the whole equations of dynamics

4.3.1. Preliminaries We now compute the base parameters $\boldsymbol{\phi}_{AB} \in \mathbb{R}^{N_{AB}}$ of the whole system from the N_J equations of motion with a forward recursive method. According to Proposition Π_ϕ for sub-chain C_j, we first choose a matrix $\boldsymbol{X}_{Aj} \in \mathbb{R}^{10(j+1)\times N_{ABj}}$ which satisfies Equation (65). Then only one matrix $\boldsymbol{Z}_{Aj} \in \mathbb{R}^{N_{ABj}\times 10(j+1)}$ satisfies

$$\dim \mathrm{V}(\boldsymbol{Y}_{Aj}) = \dim \mathrm{V}(\boldsymbol{Y}_{ABj}) = N_{ABj} \tag{65}$$

$$\boldsymbol{Y}_{Aj} = \boldsymbol{Y}_{ABj}\boldsymbol{Z}_{Aj} \tag{66}$$

By choosing $\boldsymbol{X}_{Aj}$ and $\boldsymbol{Z}_{Aj}$, the base parameters $\boldsymbol{\phi}_{AB} \in \mathbb{R}^{N_{ABj}}$ and the corresponding regressor $\boldsymbol{Y}_{ABj} \in \mathbb{R}^{6\times N_{ABj}}$ are given by

$$\boldsymbol{Y}_{ABj} \triangleq \boldsymbol{Y}_{Aj}\boldsymbol{X}_{Aj} \tag{67}$$

$$\boldsymbol{\phi}_{ABj} \triangleq \boldsymbol{Z}_{Aj}\boldsymbol{\phi}_j \tag{68}$$

4.3.2. Initialization with C_0 According to $\boldsymbol{Y}_{AO} = \boldsymbol{Y}_{O0}$ from Equations (17) and (18), and similarly to the previous case (Section 4.2), then Proposition Π_{A0} holds.

Proposition Π_{A0}. *For sub-chain C_0 made of L_0, $\boldsymbol{X}_{A0}$ and $\boldsymbol{Z}_{A0}$, defined by*

$$\boldsymbol{X}_{A0} = \boldsymbol{Z}_{A0} \triangleq \boldsymbol{E} \tag{69}$$

verify Equations (65) and (66) ($N_{OB0} = n_{B0} = 10$).

4.3.3. General case for sub-chain C_j Let us consider sub-chain C_j for $j > 0$. To compute the base parameters, we present the Proposition Π_{Aj}.

Proposition Π_{Aj}. *If Proposition Π_{Aj-1} holds, and if $\boldsymbol{X}_{j,j} \in \mathbb{R}^{10\times n_{Bj}}$ and $\boldsymbol{X}_{j,j}^{\perp} \in \mathbb{R}^{10\times(10-n_{Bj})}$ satisfy Condition Ψ_j, then the constant matrices $\boldsymbol{X}_{Aj}$ and $\boldsymbol{Z}_{Aj}$, defined by*

$$\boldsymbol{X}_{Aj} \triangleq \begin{bmatrix} \boldsymbol{X}_{Aj-1} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{X}_{j,j} \end{bmatrix} \tag{70}$$

$$\boldsymbol{Z}_{Aj} \triangleq \begin{bmatrix} \boldsymbol{Z}_{Aj-1} & \boldsymbol{Z}_{Aj-1,i}{}^{p_j}\boldsymbol{B}_j\boldsymbol{X}_{j,j}^{\perp}\boldsymbol{X}_{j,j}^{\perp\mathrm{T}} \\ \boldsymbol{O} & \boldsymbol{X}_{j,j}{}^{\mathrm{T}} \end{bmatrix} \tag{71}$$

verify Equations (65) and (66) ($N_{ABj} = N_{ABj-1} + n_{Bj}$).

Proof of Π_{Aj}. We first verify the condition when $\boldsymbol{X}_{Aj}$ does not satisfy Equation (65), such as

$$\dim(\boldsymbol{Y}_{Aj}\boldsymbol{X}_{Aj}) \neq N_{ABj} \tag{72}$$

Similarly to the proof of Π_{Oj} (Section 4.2), by using Equations (18), (28), (70), and (71), when $\exists \boldsymbol{a}_j \neq \boldsymbol{0}$, $\forall \boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\ddot{\boldsymbol{q}}$, Equation (72) is equivalent to

$$\begin{bmatrix} \boldsymbol{Y}_{ABj-1}(\boldsymbol{a}_{j-1} + \boldsymbol{Z}_{Aj-1,p_j}{}^{p_j}\boldsymbol{B}_{Aj}\boldsymbol{x}_j) + {}^0\widetilde{\boldsymbol{A}}_{Aj}{}^j\boldsymbol{Y}_j^{p_j}\boldsymbol{x}_j \\ \boldsymbol{K}_j{}^j\boldsymbol{Y}_{Aj}\boldsymbol{x}_j \end{bmatrix} = \boldsymbol{0} \tag{73}$$

where $\boldsymbol{Z}_{Aj,k} \in \mathbb{R}^{N_{OBj-1}\times 10}$ is made of the 10 columns of $\boldsymbol{Z}_{Aj}$ corresponding to link k ($0 \leq k \leq j$). Similarly to Equation (47), $\boldsymbol{Z}_{Aj,k}$ is defined as

$$\boldsymbol{Z}_{Aj,k} \triangleq \begin{bmatrix} \boldsymbol{z}_{Aj,10k+1} & \cdots & \boldsymbol{z}_{Aj,10k+10} \end{bmatrix} \tag{74}$$

The difference between Equation (73) and Equation (52) is that ${}^0\boldsymbol{A}_j$ is replaced with ${}^0\widetilde{\boldsymbol{A}}_{Aj}$ and the lower part of Equation (73) is added. Let us define ${}^0\widetilde{\boldsymbol{A}}_{ATj} \in \mathbb{R}^{6\times N_{J,j-1}}$ as

$${}^0\widetilde{\boldsymbol{A}}_{ATj} \triangleq \begin{bmatrix} {}^j\boldsymbol{A}_0 & \boldsymbol{O} & \cdots & \boldsymbol{O} \end{bmatrix} \tag{75}$$

to obtain

$${}^0\widetilde{\boldsymbol{A}}_{ATj}{}^0\widetilde{\boldsymbol{A}}_{Aj} = \boldsymbol{E} \tag{76}$$

Since $\mathrm{rank}({}^0\widetilde{\boldsymbol{A}}_{ATj}) = 6$ is obvious, then $\mathrm{rank}({}^0\widetilde{\boldsymbol{A}}_{Aj}) = 6$ from Equation (76). Therefore, we can deal with the upper part of Equation (73) in the same way as Equation (52). Hence, a necessary and sufficient condition to have Equation (72) is that $\forall k, l, \boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$ (k, $l \in [1\ n_j]$), Equations (55), (56), and

$$\boldsymbol{k}_{j,k}{}^j\boldsymbol{Y}_{Aj}\boldsymbol{x}_{j,m} = 0 \tag{77}$$

are all verified. Then $\boldsymbol{k}_{j,k}{}^j\boldsymbol{Y}_{Aj}$ can be modified as follows:

$$\boldsymbol{k}_{j,k}{}^j\boldsymbol{Y}_{Aj} = \begin{bmatrix} {}^j\ddot{\boldsymbol{p}}_j^{\mathrm{T}} & {}^j\dot{\boldsymbol{w}}_j^{\mathrm{T}} & -\frac{1}{2}{}^j\boldsymbol{w}_j^{\mathrm{T}}[{}^j\boldsymbol{w}_j\bullet] \end{bmatrix} \boldsymbol{W}_{j,k} \tag{78}$$

As can be seen from Equation (78), Equation (55) always satisfies Equation (77). Hence, if $\boldsymbol{X}_{j,j}$ satisfies Condition Ψ_j, then $\boldsymbol{X}_{Aj}$ defined by Equation (70) satisfies Equation (65).

Similarly to Section 4.2, $\boldsymbol{Z}_{Aj}$ defined by Equation (71) is constant and satisfies Equation (66). Hence, Proposition Π_{Aj} is satisfied.

4.3.4. Summary From Propositions Π_{A0} and Π_{Aj}, the following proposition holds.

Proposition 3. *For all $j \in [1\ n-1]$, if $\boldsymbol{X}_{j,j}$ satisfies Condition Ψ_j, $\boldsymbol{X}_{Aj}$ and $\boldsymbol{Z}_{Aj}$, defined by Equations (69), (70), and (71), are the composition matrices of the base parameters of the whole N_J equations of motion.*

4.4. Computation of the composition matrix of each link

Condition Ψ_j can be summarized as Equation (79), if we arrange $\boldsymbol{W}_{j,k}, \boldsymbol{W}_{j,k,l}$ ($1 \leq k \leq n_j, 1 \leq l \leq n_j$) lengthwise and compose a large matrix $\boldsymbol{W}_{Aj} \in \mathbb{R}^{(12n_j+6n_j{}^2)\times 10}$:

$$\boldsymbol{W}_{Aj}\boldsymbol{x}_{j,m}^{\perp} = \boldsymbol{0} \tag{79}$$

where $\boldsymbol{x}^{\perp}$ is the column vector of the $\boldsymbol{X}_{j,j}^{\perp}$, and $\boldsymbol{W}_{Aj}$ is the constant matrix defined by the kinematic structure of the system. Equation (79) means that the composition matrix $\boldsymbol{X}_{j,j}$ can be computed from the right-singular vectors of $\boldsymbol{W}_{Aj}$ corresponding to the non-zero singular values. Therefore, we can compose directly the base parameters from the singular value decomposition of $\boldsymbol{W}_{Aj}$.

4.5. Summary

From Propositions 2 and 3, the composition matrices $\boldsymbol{X}_{Oj}$ and $\boldsymbol{Z}_{Oj}$ of the six equations of motion of the base-link are equivalent to $\boldsymbol{X}_{Aj}$ and $\boldsymbol{Z}_{Aj}$ of the whole N_j equations of motion. Thus, we obtain the same base parameters whether using the equations of the base-link or the equations for the whole system. This proves the identifiability of the whole set of base parameters $\boldsymbol{\phi}_B$ using only the equations of motion of the base-link, i.e. Equation (10).

5. The base parameters of the floating base-link dynamics

5.1. Unidentifiable base parameters from conservation of momentum

When no external force acts on the legged system through the interaction with the environment, for example when the system is jumping or flying, the following dynamics constraint exists; the generalized forces which actuate 6 DOFs of the base-link of the system are always zero. This constraint is related to the conservation of angular momentum of the whole body, and generally cannot be integrated a second time. Owing to the non-holonomic constraint, the motion can excite all of the dynamics of the system even under the constraint. Does the non-holonomic property give some information about the inertial properties of the whole body, when we observe the motion under the constraint?

From Equation (10), the equations of motion of the base-link without external forces can be written as follows:

$$\boldsymbol{Y}_{OB}\boldsymbol{\phi}_B = \boldsymbol{0} \tag{80}$$

In the previous section, we proved that $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B$ when external forces exist. However, in the case of Equation (80), the right-hand side of Equation (80) is always zero because of the dynamics constraint, and there exists the constant vector $\boldsymbol{\phi}_B (\neq \boldsymbol{0})$ that always satisfies this linear equation. It means that the column vectors of $\boldsymbol{Y}_{OB}$ are linearly dependent, and we have

$$\dim \mathrm{V}(\boldsymbol{Y}_{\mathrm{OB}}) \leq N_B - 1 \tag{81}$$

Here $\boldsymbol{\phi}_B$ is the vector of base parameters in Equation (10), however $\boldsymbol{\phi}_B$ is not the bases in Equation (80) because of Equation (81). What is the set of non-base parameters that cannot be identified from the conservation of momentum?

Although it is difficult to derive the base parameters of general floating-base systems, we can discuss the parameters if the system satisfies the following three conditions.

Conditions

A. The system is an open-chain structure.
B. All of the joints of the system can be actuated; each element of $\boldsymbol{q}_c$, $\dot{\boldsymbol{q}}_c$, and $\ddot{\boldsymbol{q}}_c$ are independent with the others.
C. There exist at least one $\boldsymbol{q}_c$ such that (1) all of the elements of ${}^0\boldsymbol{w}_0$ are linearly independent when $\boldsymbol{q}_c$ is fixed and $\dot{\boldsymbol{q}}_c = \ddot{\boldsymbol{q}} = \boldsymbol{0}$, and (2) all the elements of ${}^0\boldsymbol{w}_0$ and $\dot{\boldsymbol{q}}_c$ are linearly independent when $\boldsymbol{q}_c$ is fixed and $\dot{\boldsymbol{q}}_c \neq \boldsymbol{0}$.

Let us define Q_c as the set of $\boldsymbol{q}_c$ satisfying Condition C.

Here we make an explanation about Condition C. For example, if the initial angular velocity of the base-link is equal to zero or the total inertia tensor around the center of total mass is constant for any joint generalized coordinates, Condition C is not satisfied. Except those cases, general multibody systems with some initial velocity satisfy Condition C.

If the system satisfies the three conditions, the following proposition holds.

Proposition 4: *If a floating multibody system satisfies Conditions A, B, and C, then* $N_B - 2 \leq \dim \mathrm{V}(\boldsymbol{Y}_{OB}) \leq N_B - 1$. *In particular, when the value of J is not constant in the range of* $\boldsymbol{q}_c \in Q_c$, *then* $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B - 1$, *where J is the trace of total inertia tensor matrix around the center of total mass.*

Let us define the following lemma that is used for the proof of Proposition 4.

Lemma 2′. *If a floating multibody system satisfies Conditions A, B, and C, then Lemma 2 holds.*

In the previous section, Lemma 2 holds under the assumption that the system can change $\boldsymbol{q}_0$, $\boldsymbol{q}_c$ and their derivatives independently. Lemma 2′ means that, if a floating system satisfies the three conditions, Lemma 2 holds even under the non-holonomic constraint. The proof of the lemmas is given in Appendix B.

5.2. Unidentifiable parameters of a floating rigid body

Before giving the proof of Proposition 4, we discuss the relation between the motion of a rigid body under the law of conservation of momentum and its inertial parameters. In this discussion, we place the origin of the coordinate system at the COM of a rigid body, and three axes of the system are chosen as principal axes inertia. We define the coordinate system as Σ_{0c}. Then, the equations of motion of a rigid body can be written as follows:

$$m^{0c}\ddot{\boldsymbol{p}}_{0c} = \boldsymbol{0} \tag{82}$$

$${}^{0c}\boldsymbol{I}_{0c}{}^{0c}\dot{\boldsymbol{\omega}}_{0c} + [{}^{0c}\boldsymbol{\omega}_{0c}\times]{}^{0c}\boldsymbol{I}_{0c}{}^{0c}\boldsymbol{\omega}_{0c} = \boldsymbol{0} \tag{83}$$

where ${}^0\boldsymbol{I} \in \mathbb{R}^{3\times3}$ is the inertia tensor expressed in Σ_0, and can be represented as the following matrix whose diagonal elements are the principal moments of inertia:

$${}^{0c}\boldsymbol{I}_{0c} \triangleq \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{ZZ} \end{bmatrix} \tag{84}$$

Equations (82) and (83) can be transformed into

$$ {}^{0c}\ddot{\boldsymbol{p}}_{0c} = \mathbf{0} \tag{85}$$

$$ {}^{0c}\dot{\boldsymbol{\omega}}_{0c} = \left[\, C_x\omega_y\omega_z \;\; C_y\omega_z\omega_x \;\; C_z\omega_x\omega_y \,\right]^{\mathrm{T}} \tag{86}$$

where ω_x, ω_y, and ω_z are the respective elements of ${}^{0}\boldsymbol{\omega}$, and C_x, C_y, and C_Z are defined as

$$ C_x \triangleq \frac{I_{yy} - I_{zz}}{I_{xx}} \tag{87}$$

$$ C_y \triangleq \frac{I_{zz} - I_{xx}}{I_{yy}} \tag{88}$$

$$ C_z \triangleq \frac{I_{xx} - I_{yy}}{I_{zz}} \tag{89}$$

The linear form with respect to the inertial parameters is as follows:

$$ {}^{0c}\boldsymbol{Y}_{0c}{}^{0c}\boldsymbol{\phi} = \mathbf{0} \tag{90}$$

If we observe the motion of a rigid body under the law of conservation of momentum, at least the following two inertial parameters cannot be identified from the observation of motion:

- the mass of one rigid body (m);
- the trace of the inertia tensor ($J \triangleq \frac{1}{2}\mathrm{tr}(\boldsymbol{I})$).

Conversely this means that from the motion we can identify the COM $\boldsymbol{s}$ and the normalized inertia tensor $\boldsymbol{I}/J$; i.e. the first- and second-order moment features computed from the continuous mass distribution describing a rigid body. This section shows the unidentifiability of the two parameters.

Proposition 5. *If the initial velocity is not zero, the vector of the initial velocity is not aligned along principal axes inertia, and all the principal moments of inertia are different, then* $\mathrm{V}({}^{0c}\boldsymbol{w}_{0c}{}^{\mathrm{T}}) = 3$ *and*

$$ \dim \mathrm{V}({}^{0}\boldsymbol{Y}) = 8 \tag{91}$$

holds. In this case, the total mass M and the trace of the inertia tensor J can be chosen as the two unidentifiable inertial parameters from Equation (90).

Proof. From Equations (85) and (86), Equation (90) can be transformed into

$$ \sum_{i=1}^{6} \Omega_i \boldsymbol{W}_i \boldsymbol{\phi} = \mathbf{0} \tag{92}$$

where $\boldsymbol{W}_i \in \mathbb{R}^{6\times 10}$ ($1 \le i \le 6$) is a constant matrix, and $\boldsymbol{\Omega}_i$ is ith element of the vector $\boldsymbol{\Omega} \in \mathbb{R}^6$ defined as

$$ \boldsymbol{\Omega} = \left[\, {\omega_x}^2 \;\; {\omega_y}^2 \;\; {\omega_z}^2 \;\; \omega_y\omega_z \;\; \omega_z\omega_x \;\; \omega_x\omega_y \,\right]^{\mathrm{T}} \tag{93}$$

As w_x, w_y, and w_z can be described as Jacobi's elliptic functions (Landau and Lifshitz, 1976), $\dim \mathrm{V}(\boldsymbol{\Omega}^{\mathrm{T}}) = 5$ holds under the following equation:

$$ \lambda_x(2E_\omega\lambda_x - ||\boldsymbol{M}_\omega||^2)\,{\omega_x}^2 + \lambda_y(2E_\omega\lambda_y - ||\boldsymbol{M}_\omega||^2)\,{\omega_y}^2 + \lambda_z(2E_\omega\lambda_z - ||\boldsymbol{M}_\omega||^2)\,{\omega_z}^2 = 0 \tag{94}$$

where E_ω is the kinetic energy of rotation, $\boldsymbol{M}_\omega \in \mathbb{R}^3$ is the vector of angular momentum, and $||\boldsymbol{M}_\omega||^2$ is constant because of the conservation of momentum. From Equation (94), ω_x^2 can be written using ω_y^2 and ω_z^2 as follows:

$$ {\omega_x}^2 = a_y{\omega_y}^2 + a_z{\omega_z}^2 \tag{95}$$

where

$$ a_y = -\frac{\lambda_y(2E_\omega\lambda_y - ||\boldsymbol{M}_\omega||^2)}{\lambda_x(2E_\omega\lambda_x - ||\boldsymbol{M}_\omega||^2)} \tag{96}$$

$$ a_z = -\frac{\lambda_z(2E_\omega\lambda_z - ||\boldsymbol{M}_\omega||^2)}{\lambda_x(2E_\omega\lambda_x - ||\boldsymbol{M}_\omega||^2)} \tag{97}$$

Then, we define $\boldsymbol{W}_A \in \mathbb{R}^{30\times 10}$ as

$$ \boldsymbol{W}_A \triangleq [\, a_y{\boldsymbol{W}_1}^{\mathrm{T}} + {\boldsymbol{W}_2}^{\mathrm{T}} \;\; a_z{\boldsymbol{W}_1}^{\mathrm{T}} + {\boldsymbol{W}_3}^{\mathrm{T}} \;\; {\boldsymbol{W}_4}^{\mathrm{T}} \;\; {\boldsymbol{W}_5}^{\mathrm{T}} \;\; {\boldsymbol{W}_6}^{\mathrm{T}} \,]^{\mathrm{T}} \tag{98}$$

As $\dim \mathrm{V}(\boldsymbol{\Omega}^{\mathrm{T}}) = 5$ holds under Equation (95), we obtain

$$ \dim \mathrm{V}({}^{0}\boldsymbol{Y}) = \mathrm{rank}(\boldsymbol{W}_A) = \mathrm{rank}(\boldsymbol{W}_Q) \tag{99}$$

where we define $\boldsymbol{W}_Q \triangleq {\boldsymbol{W}_A}^{\mathrm{T}}\boldsymbol{W}_A$. Here $\boldsymbol{W}_Q$ can be represented as

$$ \boldsymbol{W}_Q = \begin{bmatrix} 0 & \mathbf{0}^{\mathrm{T}} & \mathbf{0}^{\mathrm{T}} & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & \mathrm{diag}(\boldsymbol{w}_{q1}) & \boldsymbol{O} & \boldsymbol{O} \\ \mathbf{0} & \boldsymbol{O} & \boldsymbol{W}_{q2} & \boldsymbol{O} \\ \mathbf{0} & \boldsymbol{O} & \boldsymbol{O} & \mathrm{diag}(\boldsymbol{w}_{q3}) \end{bmatrix} \tag{100}$$

with

$$ \boldsymbol{w}_{q1} \triangleq \begin{bmatrix} 2+(1-C_y)^2+(1+C_z)^2 \\ {a_y}^2+(1+a_z)^2+(1-C_z)^2+(1+C_x)^2 \\ {a_z}^2+(1+a_y)^2+(1-C_x)^2+(1+C_y)^2 \end{bmatrix} \tag{101}$$

$$ \boldsymbol{w}_{q3} \triangleq \begin{bmatrix} 2+(1+C_y)^2+(1-C_z)^2 \\ {a_y}^2+(1-a_z)^2+(1+C_z)^2+(1-C_x)^2 \\ {a_z}^2+(1-a_y)^2+(1+C_x)^2+(1-C_y)^2 \end{bmatrix} \tag{102}$$

$$ \boldsymbol{W}_{q2} \triangleq \begin{bmatrix} 2+{C_x}^2 & C_y-C_x-1 & C_x-C_z-1 \\ C_y-C_x-1 & 2+{C_y}^2 & C_z-C_y-1 \\ C_x-C_z-1 & C_z-C_y-1 & 2+{C_z}^2 \end{bmatrix} \tag{103}$$

The matrix $\boldsymbol{W}_{q2}$ satisfies the following equation.

$$ \boldsymbol{W}_{q2}\begin{bmatrix} I_{xx} \\ I_{yy} \\ I_{zz} \end{bmatrix} \triangleq \boldsymbol{W}_{q2}\boldsymbol{\phi}_{Id} = \mathbf{0} \tag{104}$$

As all of the elements of $\boldsymbol{w}_{q1}$ and $\boldsymbol{w}_{q3}$ are positive values and $\mathrm{rank}(\boldsymbol{W}_{q2}) = 2$ holds, then we obtain

$$ \mathrm{rank}(\boldsymbol{W}_Q) = 8 \tag{105}$$

From Equations (99) and (105), we get Equation (91). From Equation (91), the following equations always hold:

$$^{0}\boldsymbol{Y}\begin{bmatrix} M \\ \mathbf{0}_3 \\ \mathbf{0}_3 \\ \mathbf{0}_3 \end{bmatrix} = \mathbf{0} \tag{106}$$

$$^{0}\boldsymbol{Y}\begin{bmatrix} 0 \\ \mathbf{0}_3 \\ \boldsymbol{\phi}_{Id} \\ \mathbf{0}_3 \end{bmatrix} = \mathbf{0} \tag{107}$$

where $\mathbf{0}_3$ is the 3×1 zero vector. Equations (106) and (107) imply the norms of the two vectors are unidentifiable from Equation (90). These two vectors can be normalized by the mass M and the trace of the inertia tensor J, which are the physical quantities that are a linear combination of inertial parameters and are invariant under coordinate transformations. Hence, we can represent $^{0}\boldsymbol{\phi}$ using the two normalized vectors $\boldsymbol{e}_M$ and $\boldsymbol{e}_I \in \mathbb{R}^{10}$ as

$$^{0}\boldsymbol{\phi} \triangleq M\boldsymbol{e}_M + J\boldsymbol{e}_I \tag{108}$$

The space spanned by $\boldsymbol{e}_M$ and $\boldsymbol{e}_I$ is equivalent with the null-space of $V(^{0}\boldsymbol{Y})$, and from the computation of the null-space of $V(^{0}\boldsymbol{Y})$, we can identify $\boldsymbol{e}_M$ and $\boldsymbol{e}_I$.

We can make the same discussion in the arbitrary coordinate system Σ. Equation (90) can be transformed in Equation (109), if using the coordinate transformation matrix $\boldsymbol{B}_0 \in \mathbb{R}^{10\times 10}$ of the inertial parameters from Σ to Σ_0, and $\boldsymbol{A}_0 \in \mathbb{R}^{6\times 6}$ of the forces and moments actng on the rigid body from Σ to Σ_0. The detailed computation of both matrices is shown in Section 3.

$$\boldsymbol{Y}\boldsymbol{\phi} = \mathbf{0} \tag{109}$$

with $\boldsymbol{Y} = \boldsymbol{A}_0{}^{0}\boldsymbol{Y}\boldsymbol{B}_0{}^{-1}$ and $\boldsymbol{\phi} = \boldsymbol{B}_0{}^{0}\boldsymbol{\phi}$. As $\boldsymbol{A}_0$ and $\boldsymbol{B}_0$ are constant regular matrices, then it is clear that $\dim \mathrm{V}(\boldsymbol{Y}) = \dim \mathrm{V}(^{0}\boldsymbol{Y}) = 8$. Since M and J are invariant under coordinate transformations, we can consider the unidentifiable parameters from Equation (90) as M and J in the same way.

5.3. Proof of Proposition 4

Equation (80) can be transformed as follows:

$$\mathbf{0} = \sum_{j=0}^{N_L} {}^{0}\boldsymbol{A}_j{}^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{\phi}_{Aj} \tag{110}$$

where $\boldsymbol{\phi}_{Aj} \in \mathbb{R}^{10}$ is defined as the following vector:

$$\boldsymbol{\phi}_{Aj} \triangleq \boldsymbol{\phi}_j + \sum_{k\in\mathcal{J}_j} {}^{j}\boldsymbol{B}_k\boldsymbol{\phi}_k \tag{111}$$

Here $\boldsymbol{\phi}_{Aj}$ corresponds to the total inertial parameters of the sub-chain consisting of L_j and $\mathcal{J}_j$ that is the whole set of children of L_j.

First, we assume that all of the values of $\boldsymbol{q}_c$, $\dot{\boldsymbol{q}}_c$, $\ddot{\boldsymbol{q}}_c$ are fixed. The fixed value of $\boldsymbol{q}_c$ is defined as $\boldsymbol{q}_{c,fix}$ The velocity and acceleration of the base-link obey the non-holonomic constraint. Then, Equation (110) can be written as

$$^{0}\boldsymbol{Y}_0^{p_0}\boldsymbol{\phi}_{A0} = \mathbf{0} \tag{112}$$

As all of the joints are fixed, we can treat the system as a single rigid body. In this case, each element of $^{0}\boldsymbol{w}_0$ is linearly independent from Condition C. From Proposition 5, M and $J_{fix}(\triangleq \frac{1}{2}\mathrm{tr}(\boldsymbol{J}_{fix}))$ are not identifiable, where M is the total mass of the system and $\boldsymbol{J}_{fix}$ is the total inertia tensor matrix at $\boldsymbol{q}_c = \boldsymbol{q}_{c,fix}$. Then, $\boldsymbol{\phi}_{A0}$ can be written as

$$\boldsymbol{\phi}_{A0} = M\boldsymbol{e}_{M,fix} + J_{fix}\boldsymbol{e}_{J,fix} \tag{113}$$

where $\boldsymbol{e}_{M,fix} \in \mathbb{R}^{10}$ and $\boldsymbol{e}_{J,fix} \in \mathbb{R}^{10}$ are the normalized vectors corresponding to M and J_{fix}, respectively.

Next, $\boldsymbol{q}_c$ remains fixed to $\boldsymbol{q}_{c,fix}$, but $\dot{\boldsymbol{q}}_c$, $\ddot{\boldsymbol{q}}_c$ are changed independently. If we change only $\dot{\boldsymbol{q}}_j$ and $\ddot{\boldsymbol{q}}_j$ of the jth joint, and if the velocities and accelerations of other joints are zero, then Equation (110) becomes as follows:

$$^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{\phi}_{Aj} = {}^{0}\boldsymbol{y}_{M,fix}M + {}^{0}\boldsymbol{y}_{J,fix}J_{fix} \tag{114}$$

where $^{0}\boldsymbol{y}_{M,fix} \triangleq {}^{j}\boldsymbol{A}_0{}^{0}\mathrm{y}_0^{p_0}\boldsymbol{e}_{M,fix}$ and $^{0}\boldsymbol{y}_{J,fix} \triangleq {}^{j}\boldsymbol{A}_0{}^{0}\boldsymbol{Y}_0^{p_0}\boldsymbol{e}_{J,fix}$.

Here, when $\boldsymbol{q}_c = \boldsymbol{q}_{c,fix}$ and $\dot{\boldsymbol{q}}_i = \ddot{\boldsymbol{q}}_i = \mathbf{0}$ $(i \neq j)$, we discuss a necessary and sufficient condition such that $\forall\dot{\boldsymbol{q}}_j, \ddot{\boldsymbol{q}}_j$, there exists a constant vector $\boldsymbol{x}_j(\neq 0) \in \mathbb{R}^{10}$ that always satisfies

$$^{j}\boldsymbol{Y}_j^{p_j}\boldsymbol{x}_j = \mathbf{0} \tag{115}$$

From Condition C and Lemma 2′, the necessary and sufficient condition that there exist $\boldsymbol{x}_j$ satisfying Equation (115) is that there exist $\boldsymbol{x}_j$ satisfying Equation (55) and (56). From Condition Ψ_j, $\dim \mathrm{V}(^{j}\boldsymbol{Y}_j^{p_j}) = n_{B,j}$ is verified even when $\boldsymbol{q}_c$ is fixed to $\boldsymbol{q}_{c,fix}$. As we can have the same discussion for all joints, the number of base parameters of each link except the base-link is not changed if using the model of Equation (80). This leads to $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) \geq N_B - 2$, and all of the base parameters can be represented by two unknown parameters M and J_{fix}.

We have proven that $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B - 2$ even when $\boldsymbol{q}_c$ is fixed to $\boldsymbol{q}_{c,fix}$. In this case, the base parameters of all links can be represented by the total mass M, the total inertia tensor J_{fix} for $\boldsymbol{q}_c = \boldsymbol{q}_{c,fix}$, and the two normalized bases $\boldsymbol{e}_{B,M}, \boldsymbol{e}_{B,J} \in \mathbb{R}^{N_B}$ in the null space of $\mathrm{V}(\boldsymbol{Y}_{OB})$:

$$\boldsymbol{\phi}_B = \boldsymbol{e}_{B,M}M + \boldsymbol{e}_{B,J}J_{fix} \tag{116}$$

If we assume $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B - 2$ for any arbitrary generalized coordinates, it is clear that $\boldsymbol{e}_M$ and $\boldsymbol{e}_J$ are also the bases of the null space of $\mathrm{V}(\boldsymbol{Y}_{OB})$, and the base parameters can be written as the same way as

$$\boldsymbol{\phi}_B = \boldsymbol{e}_{B,M}M + \boldsymbol{e}_{B,J}J \tag{117}$$

From Equation (116) and (117), we obtain $J = J_{fix}$ for all $\boldsymbol{q}_c \in Q_c$, as the total mass M is constant. Thus, if $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B - 2$, then J is constant for all $\boldsymbol{q}_c \in Q_c$. This leads to the fact that if there exists $\boldsymbol{q}_c \in Q_c$ such that $J \neq J_{fix}$, then $\dim \mathrm{V}(\boldsymbol{Y}_{OB}) = N_B - 1$.

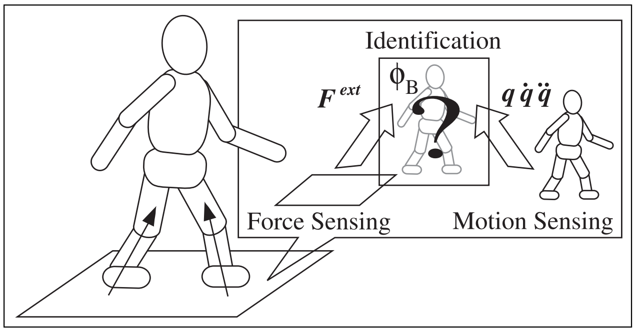


Fig. 4. Conceptual diagram of the proposed identification method based on external force sensing.

5.4. Summary

If a floating-base system satisfies Conditions A, B, and C, the regressor matrix $\boldsymbol{Y}_{OB}$ of Equation (81) satisfy dim V($\boldsymbol{Y}_{\mathrm{OB}}$) $\geq$ $\mathrm{N_B} - 2$, and the unidentifiable parameters from Equation (81) can be regarded as the total mass M and J that is the trace of the inertia tensor matrix around the center of total mass. In particular, when the value of J is not constant in the range of $\boldsymbol{q}_c \in Q_c$, dim V($\boldsymbol{Y}_{\mathrm{OB}}$) $= \mathrm{N_B} - 1$ is verified, and the unidentifiable parameter is only the total mass M.

The general open kinematic chain with no unactuated joint can satisfy Conditions A, B, and C, if the base-link of the system starts the motion with some initial velocity. In many cases, the trace of the total inertia tensor around the center of total mass is variable when the joints move, and thus dim V($\boldsymbol{Y}_{OB}$) $= N_B - 1$. Since $\boldsymbol{Y}_{OB}\boldsymbol{\phi}_B = \boldsymbol{0}$ always holds and the dimension number of the null space of V($\boldsymbol{Y}_{OB}$) is only one, the basis of the null space is equal to the normalized vector of the base parameters: $\boldsymbol{\phi}_{B,e} \triangleq \boldsymbol{\phi}_B/||\boldsymbol{\phi}_B||$. This means that if we observe the motion of the floating system in the conservation of momentum, $\boldsymbol{\phi}_{B,e}$ can be identified without the information of forces.

6. Identification of legged mechanisms based on base-link dynamics

6.1. The implementation method using force sensors

The equations of motion of legged systems are given by Equation (9). This represents a minimal identification model. Most common identification methods use Equation (9) to identify $\boldsymbol{\phi}_B$. Thus, those methods require the following information:

- the generalized coordinates $\boldsymbol{q}_0$ of the base-link;
- the joint information $\boldsymbol{q}_c$ and $\boldsymbol{\tau}$;
- the external forces $\boldsymbol{F}_k^{ext}$ for the N_c contact points.

Joint torque can either be measured directly by a torque sensor or calculated from the actuator model. However, humanoids are rarely equipped with torque sensors in their limbs, and their actuators dynamics has many nonlinear internal components such as friction and viscosity of the joints. As there are many joints to be measured, these problems make it difficult to identify the parameters of humanoids.

In previous sections, it has been proven that the reduction of the system to these six equations of the base-link keeps the number of base parameters that are structurally identifiable unchanged. Then, let us perform the identification of $\boldsymbol{\phi}_B$ using only the upper-part of Equation (9), i.e. the equations of motion of the base-link shown in (10). Only measurement of the contact forces, the joint angles, and the generalized coordinates are required. The information can be measured, for example, by six-axis force/moment sensors, encoders, gyroscopes, and acceleration sensors. In addition, by measuring directly the contact force $\boldsymbol{F}^{ext}$ there is no need to discriminate between the phases of double support and single support. It should be noted that the method does not depend on whether the legged system is slipping on the ground or not, since the friction forces are also exerted on the system as the external forces and need not be handled separately. The identification of the dynamics can be performed all along the motion whatever the type of support phase. As we only use the six equations of the base-link, the computational cost is quite low. The fast computational speed is useful for application to real-time identification or adaptive control (Slotione and Li, 1991). The conceptual diagram of this method is presented in Figure 4.

Equation (10) can thus be considered as a classical problem of identification. For the identification process, we have to compute the regressor matrix $\boldsymbol{Y}_{OB}$ and the total six-axis external forces $\boldsymbol{F}(\triangleq \sum_{k=1}^{N_c} \boldsymbol{J}_{Ok}{}^{\mathrm{T}}\boldsymbol{F}_k^{ext})$ acting on the base-link at every sampling time, measuring generalized coordinates, joint torques, and external forces acting on a robot. Then, we collect sampled regressors and forces of Equation (10) at $t = t_1, t_2, \ldots$ lengthwise. After sampling along a motion, the problem can be solved by the least-squares method as follows:

$$\widehat{\boldsymbol{\phi}}^B = \boldsymbol{\mathcal{Y}}_{B,n_T}{}^{-1}\boldsymbol{\mathcal{F}}_{n_T} \tag{118}$$

where:

- $\widehat{\boldsymbol{\phi}}_B$ is the vector of identified base parameters;
- $\boldsymbol{\mathcal{Y}}_{B,n} \in \mathbb{R}^{N_B \times N_B}$, $\boldsymbol{\mathcal{F}}_n \in \mathbb{R}^{N_B}$ are defined as follows:

$$\boldsymbol{\mathcal{Y}}_{B,n} = \sum_{i=1}^{n}(\boldsymbol{Y}_{OB,i}^{\mathrm{T}}\boldsymbol{W}_F\boldsymbol{Y}_{OB,i}) \tag{119}$$

$$\boldsymbol{\mathcal{F}}_n = \sum_{i=1}^{n}(\boldsymbol{Y}_{OB,i}^{\mathrm{T}}\boldsymbol{W}_F\boldsymbol{F}_i) \tag{120}$$

- n_T is the number of samples;
- $\boldsymbol{Y}_{OB,i}$ and $\boldsymbol{F}_t$ are the ith sampled values of $\boldsymbol{Y}_{OB}$ and $\boldsymbol{F}$, respectively;
- $\boldsymbol{W}_F \in \mathbb{R}^{6\times 6}$ is the weight matrix of external forces.

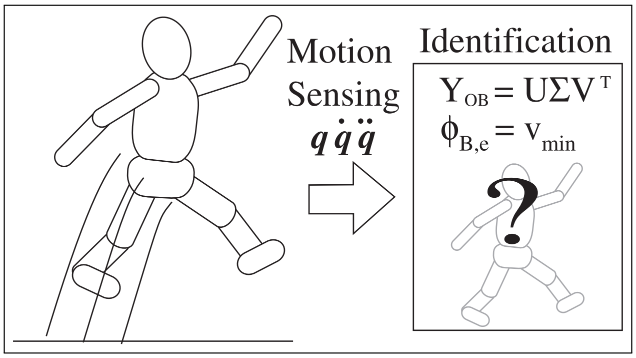


Fig. 5. Overview using free-base dynamics.

6.2. The implementation method using the floating-base dynamics

In this section we discuss the special case when no external force acts on the legged system through the interaction with the environment: when the system is jumping or flying. In that case, the equations of motion of the base-link are written as Equation (80). As we can assume that the total external forces are equal to zero, we do not need to use the force sensors.

Let us assume that the system both satisfies the three conditions shown in Section 5 and can change the value of the trace of the total inertia tensor around the center of total mass. In that case, the basis of the null space is equal to the normalized vector of the base parameters: $\boldsymbol{\phi}_{B,e}$. It should be noted that if the system can change the linear and angular momentum of the base-link through the contact with the environment such as the jumping legged systems, all three conditions are satisfied. Moreover, the general legged system can change the value of the trace of the total inertia tensor except for some special cases such that the system consists of only one rigid body.

Actually for the identification, the parameters cannot be identified by the standard least-squares method from Equation (118) because $\text{rank}(\boldsymbol{\mathcal{Y}}_{B,n_T}) < N_B$ always holds. In this section, the method to identify the unit vector $\boldsymbol{\phi}_{B,e}$ using the total least-squares method (Golub and Van Loan, 1980) is presented.

The regressor $\boldsymbol{\mathcal{Y}}_{B,n_T}$ computed from the measured motion information and the geometric parameters of the system includes the measurement noise and the geometric modeling error. Let us define $\Delta\boldsymbol{\mathcal{Y}}_{B,n_T}$ as error of the regressor, and $\widehat{\boldsymbol{\mathcal{Y}}}_{B,n_T}$ as the measured regressor. Then, $\widehat{\boldsymbol{\mathcal{Y}}}_{B,n_T}$ can be expressed by

$$\widehat{\boldsymbol{\mathcal{Y}}}_{B,n_T} = \boldsymbol{\mathcal{Y}}_{B,n_T} + \Delta\boldsymbol{\mathcal{Y}}_{B,n_T} \tag{121}$$

The singular value decomposition (Golub and Van Loan, 1983) of the measured regressor $\widehat{\boldsymbol{\mathcal{Y}}}_{B,n_T}$ is given by

$$\widehat{\boldsymbol{\mathcal{Y}}}_{B,n_T} = \begin{bmatrix} \boldsymbol{U} \ \boldsymbol{u}_{min} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma} & \boldsymbol{0} \\ \boldsymbol{0}^{\mathrm{T}} & \sigma_{min} \end{bmatrix} \begin{bmatrix} \boldsymbol{V} \ \boldsymbol{v}_{min} \end{bmatrix}^{\mathrm{T}} \tag{122}$$

Then, we identify the singular vector corresponding to the smallest singular value σ_{min} as the estimated vector $\widehat{\boldsymbol{\phi}}_{B,e}$.

$$\widehat{\boldsymbol{\phi}}_{B,e} = \boldsymbol{v}_{min} \tag{123}$$

Here $\widehat{\boldsymbol{\phi}}_{B,e}$ minimizes the Frobenius norm $||\Delta\boldsymbol{\mathcal{Y}}_{B,n_T}||_F = \sigma_{min}$. The conceptual diagram of this identification method is presented in Figure 5.

In order to identify $\boldsymbol{\phi}_B$, we need at least one base parameter. Here, we show the example to identify $\boldsymbol{\phi}_B$ using the total mass of the system, because the total mass can be measured precisely. When we compose the base parameters, we can choose one of them as the total mass. We define the parameter in $\widehat{\boldsymbol{\phi}}_{B,e}$ corresponding to the total mass as $\widehat{M}_e$, and the measured total mass as $\widehat{M}$. Then we can identify $\widehat{\boldsymbol{\phi}}_B$ as follows:

$$\widehat{\boldsymbol{\phi}}_B = \frac{\widehat{M}}{\widehat{M}_e} \widehat{\boldsymbol{\phi}}_{B,e} \tag{124}$$

The unidentifiability of $||\boldsymbol{\phi}_B||$ derives from the lack of information about unit system of force; we can know the relative ratio among the internal forces generated in each joint, while we cannot compare the internal forces with the external ones acting on the system.

In the field of the identification of space robots, Murotsu et al. (1994) proposed a method to identify the inertial parameters of a robot and an object handled by the robot. They assumed that the parameters of the base body system are known. And they identify the unknown parameters from the known dynamics. In the sense of using known parameters, the method is similar to theirs. However, we can identify the unit vector of the base parameters without known parameters or force information. If we want to identify the normal base parameters, we can identify from at least one known base parameter.

The features of the method are summarized as follows:

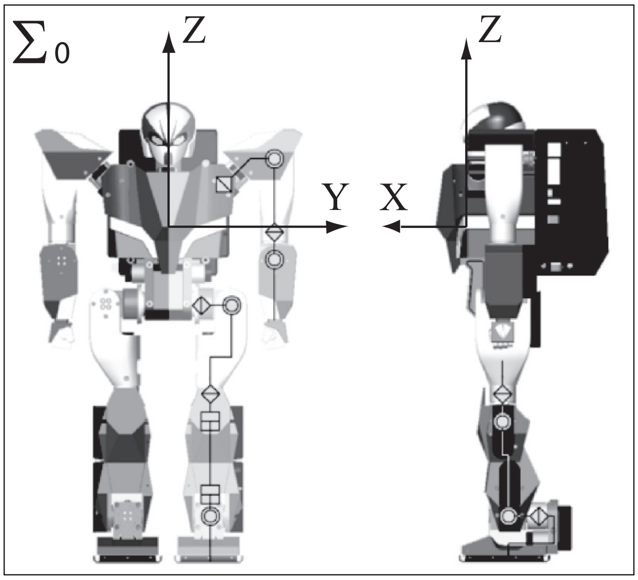
- the method can be applied when the dimension number of the regressor decreases by only one; general legged systems can satisfy the assumption;
- only the motion under the law of conservation of momentum is measured for identification; the motion can be measured by both internal sensors and external ones;
- the normalized parameters $\boldsymbol{\phi}_{B,e}$ can be identified without force measurement, only from motion data.

7. Application to a small-size humanoid robot

7.1. Experimental setup

The method using force sensors shown in Section 6.1 was tested on a small size humanoid robot UT-μ2 (Sugihara et al., 2005). The structure and specifications of the robot are shown in Figure 6.

The inertial parameters of each link are approximately known from the CAD data of the robot, however cables and some additional parts of hardware are not taken into account which is critical for a robot of those dimensions. The total mass of the robot computed from the CAD data is 6.9 kg; however, in fact, the measured total mass is 8.0



Name:	UT-μ2:magnum
Height:	540 [mm]
Weight:	8.0 [kg]
Number of joints:	20 (12 for legs, 8 for arms)

Fig. 6. Kinematic structure and specifications of the small-size humanoid robot.

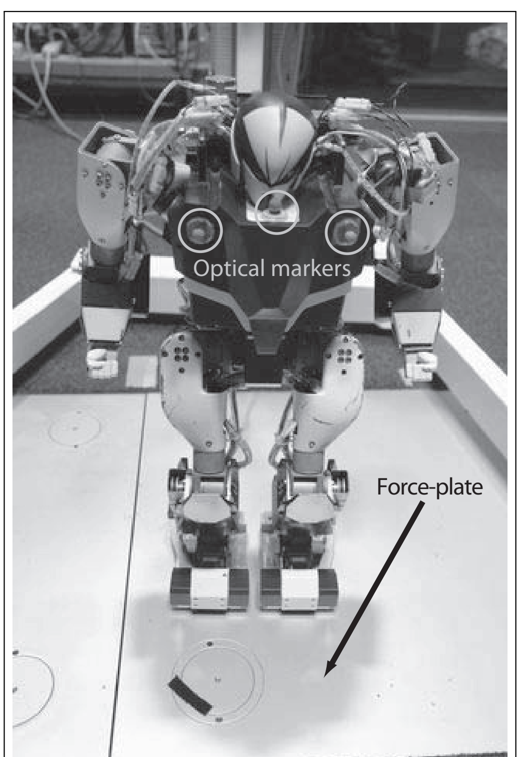


Fig. 7. UT-μ2 during the experiments: equipped with optical markers for motion capture and on the force plate.

kg. The error of 14% has an impact on most of the other dynamics parameters. The identification is thus essential.

The six components of the external forces and moments were measured by the force-plate (Kistler). The resolutions of each force-plate are ± 0.7 N of forces and ± 0.3 Nm of moments, and the drifts are ± 0.3 N/min of forces and ± 0.1 Nm/min of moments. Since the robot is not equipped with a gyroscope, the generalized coordinates $\boldsymbol{q}_0$ were measured by a motion capture system. The base-link of the robot was equipped with optical markers as shown in Figure 7. They are recorded by a commercial optical motion capture system consisting of 10 cameras (Motion Analysis). The resolution of the capture volume is about 3 mm. Then q_0 was reconstructed from the motion data. The joint angles were measured by the joint encoders. Motion data, force data, and robot data were synchronized.

Although the whole-base parameters can be theoretically identified in the method, we focused on the inertial parameters of the legs for all of the experiments in this section; the upper body was fixed in a pose, and only motions of the legs and the trunk were used. If identifying the whole parameters, the estimation errors are expected to increase, because the ratio of mass of the arms and that of the legs is about 1:3, and the arms have small parameters that cannot be measured with respect to the sensors resolution. In this case, the robot has 12 DOFs and 13 links; the trunk is chosen as the base-link and the robot has 6 rotational joints in each leg.

The method to compose the base parameters $\boldsymbol{\phi}_B$ and the regressor $\boldsymbol{Y}_{OB}$ of the robot is now detailed. If the z-axis of the coordinate system of each joint is aligned along each joint axis, the composition matrix $\boldsymbol{X}_{j,j}(1 \leq j \leq 12)$ of each link except for the base-link can be chosen as follows:

$$\boldsymbol{X}_{j,j} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{125}$$

From Equations (38), (39), (40), and (125), we can compute both $\boldsymbol{X}_O$ and $\boldsymbol{Z}_O$, and then can compose the base parameters and the regressor by using (36) and (37). The number of the base parameters of the robot is $94(= 10 + 7 \times 12)$.

7.2. Motions for identification

Once modeling and measuring issues are resolved, we have to adequately choose the motions that can excite the whole dynamics: PE trajectories. These motions are usually chosen from considerations on the condition number of the regressor (Gautier and Khalil, 1992); in general, the closer to 1 the condition number of the regressor is, the better excited the dynamics is. For the identification of humanoid robots, it is usually difficult to generate the PE trajectories because of the number of DOFs of them and the stability problem under the non-holonomic properties of their dynamics. In those cases, some feasible motions are often measured to find the optimal combination of them for identification. Therefore, in the experiment, several motions were generated as follows:

1. walking four steps forward (*walk*);
2. taking four side steps to the right (*side step*);
3. bowing (*bow*);
4. squat (*squat*);
5. taking a step with the left foot and squat (*left squat*);
6. taking a step with the right foot and squat (*right squat*);
7. bending the trunk in all directions and twisting it to the left and right (*bend*).

The condition number of the regressor obtained with each motion is shown in Table 1. The motion with low condition number is considered to be a PE trajectory. For example, the condition numbers of *bow*, *squat*, and *bend* were relatively large, since the robot did not move its feet in those motions and thus could not excite the whole dynamics. Among the seven motions, *walk* had the smallest condition number and could be considered as the best motion if only one motion is used for identification. The best combination of motions from the whole set that minimize the condition number of the combined regressor is the combination of *walk*, *side step*, and *right squat*; the motion set guarantees the lowest condition number cond($\boldsymbol{Y}_{OB}$) $= 312$, since the robot can move its total COM three-dimensionally and can excite the whole dynamics. It should be noted that, if we use all of the equations corresponding to the joints, the condition number of the combined regressor itself can be reduced. Although the joint torque can be calculated from the actuator models, the actuator dynamics has many nonlinear internal components. These problems make it difficult to identify the parameters of humanoids.

Table 1. Condition number of the regressor for seven types of motions.

Motion	Walk	Side step	Bow	Squat
cond($\boldsymbol{Y}_{OB}$)	496	885	4.39×10^{10}	1.65×10^{10}
Motion	Left squat	Right squat	Bend	
cond($\boldsymbol{Y}_{OB}$)	1810	1460	6810	

As mentioned above, in the identification of legged systems, it is usually difficult to generate the PE trajectories. In particular, in the case of human identification, it is not feasible to generate precise prescribed motions. In the same way as in this section, it is reasonable to select the PE trajectories from a set of recorded data (Venture et al., 2009a).

7.3. Results

As described in the previous section, the concatenation of *walk*, *side step*, and *right squat* was used for identifying the parameters of the robot. If the ground truth parameters of the system are unknown, the reconstructed forces $\boldsymbol{Y}_{OB}\widehat{\boldsymbol{\phi}}_B$ estimated from the identified parameters $\widehat{\boldsymbol{\phi}}_B$ are often compared with the measured ones $\boldsymbol{F}^{ext}$ to evaluate the results of identification. We generally use motions that are not used during the identification procedure for the comparison (i.e. cross-validation). Therefore, the remaining recorded motions are used for cross-validations.

The obtained results are presented in Figure 8 and Figure 9 for cross-validation using *left squat* and *bending*, respectively. They give the comparison of the vector of total external forces $\boldsymbol{F}^{ext}$ acting on the base-link (gray thick lines) obtained from the force plate data, with the reconstructed vector of forces $\boldsymbol{Y}_{OB}\widehat{\boldsymbol{\phi}}_B$ (black thin lines) computed from the identified parameters $\widehat{\boldsymbol{\phi}}_B$, for each of the six components. In addition, they also show the reconstructed vector of forces $\boldsymbol{Y}_{OB}\boldsymbol{\phi}_{B_{apriori}}$ (black thin dotted line) estimated from the *a priori* parameters $\boldsymbol{\phi}_{B_{apriori}}$ obtained from the CAD data. The absolute mean errors of six components in the case using the CAD data and in the case using the identified parameters are shown respectively in the captions.

The (94×1) vector of base parameters $\widehat{\boldsymbol{\phi}}_B$ was fully estimated. As can be seen from the cross-validation figures, the estimated parameters allowed a good prediction of the external forces, even with a motion completely different from the ones used for the identification. It should be noted that there was a clear difference when using the estimated parameters (black thin lines) or the *a priori* ones (black thin dotted lines). With the CAD data, the vertical force Fz had an error of almost 13 N due to the error on the *a priori* mass. There was also an error of 1.8 Nm on the sagittal moment Ny. Locally the moment Nx presented an error of 0.9 Nm. On the other hand, the reconstructed forces with the estimated parameters were obviously more accurate. The identified total mass is 7.93 kg; the error from the measured value is 0.07 kg. As can be seen from Figure 8 and Figure 9, the mean errors of six components estimated from the identified parameters are within the resolutions of the force plates.

8. Application to two floating systems without force measurement

8.1. Two links connected with a rotational joint

The method using only the motion data under the law of conservation of momentum shown in Section 6.2 was tested on two examples: a simple chain made of two links and the human body dynamics. In this section we show that the proposed identification method is still valid when the legged system is jumping or there are no force sensors. For practical use, the ground force measurement should be needed if accurate sensors are available.

First, this section shows the identification results of a simple two-link system as shown in Figure 10. Two links are connected with a single rotational joint, which has no

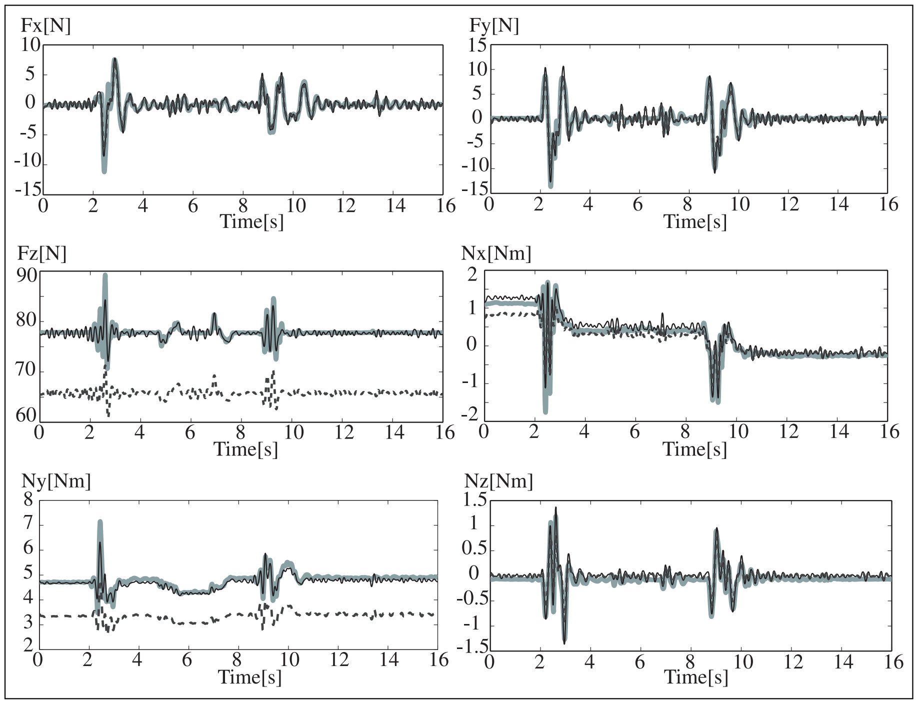


Fig. 8. Cross-validation of the external force for *left squat* from the identification result of the three motions (*walk*, *side step*, and *right squat*). The gray thick lines show the measured forces, the black thin lines indicate the estimated forces from the identified parameters, and the black thin dotted line represent the estimated forces from the CAD data. The mean errors of six-axis forces in the two cases are as follows: (CAD data) Fx 0.41[N], Fy 0.56[N], Fz 12.01[N], Nx 0.13[Nm], Ny 1.42[Nm], Nz 0.08[Nm]; (Identified parameters) Fx 0.46[N], Fy 0.63[N], Fz 0.46[N], Nx 0.11[Nm], Ny 0.11[Nm], Nz 0.08[Nm].

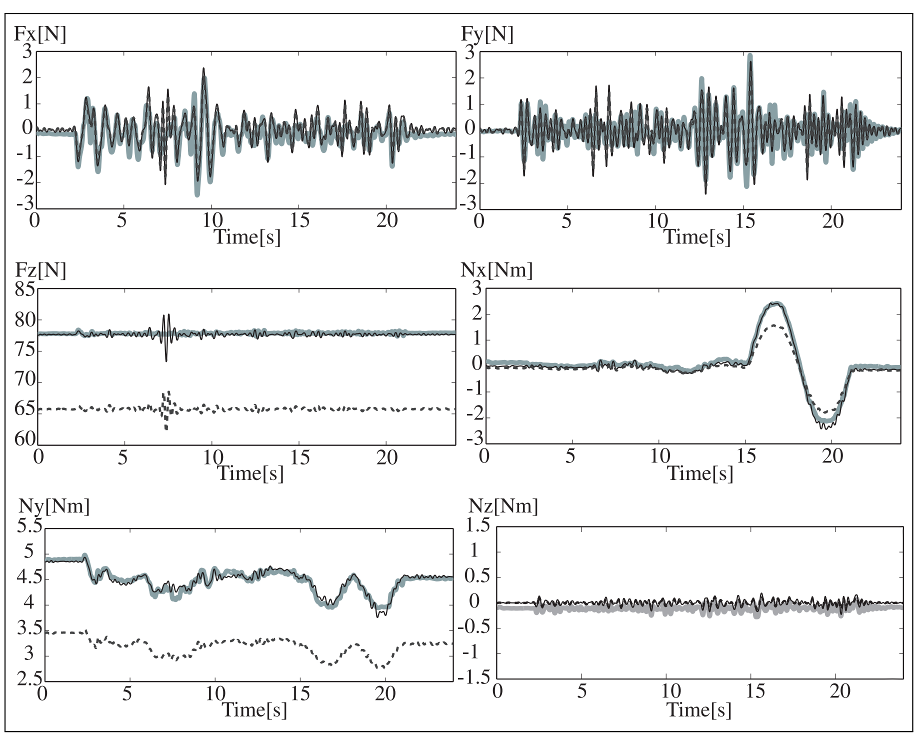


Fig. 9. Cross-validation of the external force for *bending* from the identification result of the three motions (*walk*, *side step*, and *right squat*). The three types of lines are used in the same manner as Figure 8. The mean errors of six-axis forces in the two cases are as follows: (CAD data) Fx 0.25[N], Fy 0.25[N], Fz 12.09[N], Nx 0.22[Nm], Ny 1.29[Nm], Nz 0.11[Nm]; (Identified parameters) Fx 0.26[N], Fy 0.27[N], Fz 0.29[N], Nx 0.10[Nm], Ny 0.05[Nm], Nz 0.11[Nm].

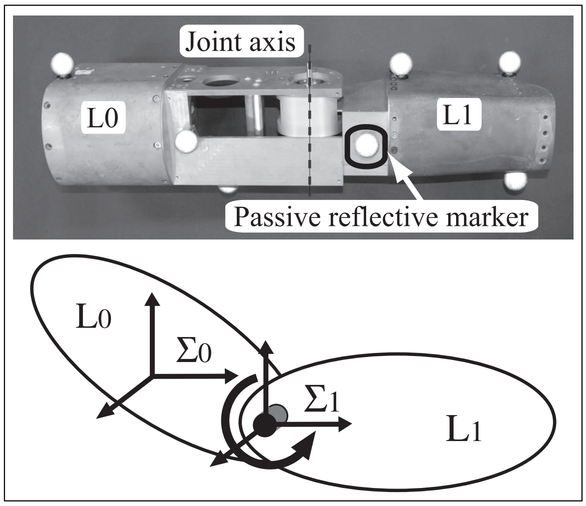


Fig. 10. Kinematic chain made of two links connected with a rotational joint.

Table 2. Identification results with experimental data when the chain made of two links connected with a rotational joint is flying freely without external forces.

Link ID	name	Measured parameter	Identified parameter
L0	$\phi_{Be,0,1}$	0.997	0.997
	$\phi_{Be,0,2}$	0.000	–0.002
	$\phi_{Be,0,3}$	0.064	0.064
	$\phi_{Be,0,4}$	0.006	0.003
	$\phi_{Be,0,5}$	0.013	0.006
	$\phi_{Be,0,6}$	0.008	0.003
	$\phi_{Be,0,7}$	0.019	0.017
	$\phi_{Be,0,8}$	0.000	0.003
	$\phi_{Be,0,9}$	0.000	0.003
	$\phi_{Be,0,10}$	0.000	0.000
L1	$\phi_{Be,1,1}$	–0.036	–0.029
	$\phi_{Be,1,2}$	–0.003	–0.006
	$\phi_{Be,1,3}$	–0.005	–0.008
	$\phi_{Be,1,4}$	0.006	0.001
	$\phi_{Be,1,5}$	0.000	–0.004
	$\phi_{Be,1,6}$	0.000	0.001
	$\phi_{Be,1,7}$	0.000	–0.001

Table 3. Chosen DOFs in the model of the human body.

Name of joint	Type of joint	Number of DOFs
Neck	Spherical	3
Waist	Spherical	3
Right shoulder	Spherical	3
Right elbow	Revolute	1
Right wrist	Spherical	3
Left shoulder	Spherical	3
Left elbow	Revolute	1
Left wrist	Spherical	3
Right hip	Spherical	3
Right knee	Revolute	1
Right ankle	Spherical	3
Left hip	Spherical	3
Left knee	Revolute	1
Left ankle	Spherical	3

actuator and whose axis is simply supported by bearings. For identification, the links were thrown into the air several times. The motion in the air was recorded with the motion capture system mentioned in the previous section. The links were equipped with optical markers as shown in Figure 10, and the motion were totally recorded for about 20 seconds. From the measured position of markers, the generalized coordinates $\boldsymbol{q}_0$ of the base-link and the joint angle $\boldsymbol{q}_c$ were computed. The derivatives $\dot{\boldsymbol{q}}_0$, $\dot{\boldsymbol{q}}_c$, $\ddot{\boldsymbol{q}}_0$, $\ddot{\boldsymbol{q}}_c$ were also obtained numerically. Here $\boldsymbol{Y}_{all}$ was also computed using those variables and the geometric parameters of the system at every sampling time.

This system has two links and one rotational joint. Thus, the total number of base parameters $\boldsymbol{\phi}_B$ of the system is 17; 10 of the parameters $\boldsymbol{\phi}_{B0}$ of L0, and 7 of the parameters $\boldsymbol{\phi}_{B1}$ of L1, where the notation L0 is defined as the base-link of the system, and L1 as the other side. The z-axis of the coordinate system of L1 is aligned along the joint axis. The base parameters $\boldsymbol{\phi}_B = [\boldsymbol{\phi}_{B0}{}^{\mathrm{T}}\ \boldsymbol{\phi}_{B1}{}^{\mathrm{T}}]^{\mathrm{T}}$ of this system can be computed from Equations (38), (39), (40), and (125).

The joint of this system is underactuated, so the system does not satisfy Condition B shown in Section 5. The numerical identifiability (Gautier, 1990) of the system was checked from the dynamics simulation. With the model whose inertial parameters are equal to the measured ones, the motion was simulated without actuation of the joint and external forces for 5 seconds. The initial joint angle was chosen when the joint is extended, and all of the generalized velocities were initialized at 1. The obtained maximum singular value of $\mathcal{Y}_{B,n_T}$ was 1.26×10^2, the smallest singular value was 7.92×10^{-6}, and the second smallest was 1.09×10^0. It is clear that the decrease of the dimension of $\mathrm{V}(\mathcal{Y}_{B,n_T})$ is only one, so $\boldsymbol{\phi}_{B,e}$ of this system is numerically identifiable.

The identified unit vector $\widehat{\boldsymbol{\phi}}_{B,e}$ of base parameters and the measured unit vector $\boldsymbol{\phi}_{B,e}$ are shown in Table 2. In order to obtain $\boldsymbol{\phi}_{B,e}$, the weight of all of the components of each link was measured. The COM and the inertia tensor were calculated using CAD data. As it can be seen from Table 2, the identified parameters were matching with the measured data, especially large inertial parameters such as mass and COM. Each absolute error of a parameter was under 0.007, so the parameters were successfully identified. However, the maximum singular value of the measured regressor matrix was 1191, the smallest singular value is 18, and the second smallest was 76, so the second smallest value was near the smallest one. This can be explained not only by measurement errors, numerical computation of derivatives, model errors, and existence of air resistance, but also by the poor excitation property of the motion in the air mainly caused by lack of actuated joint.

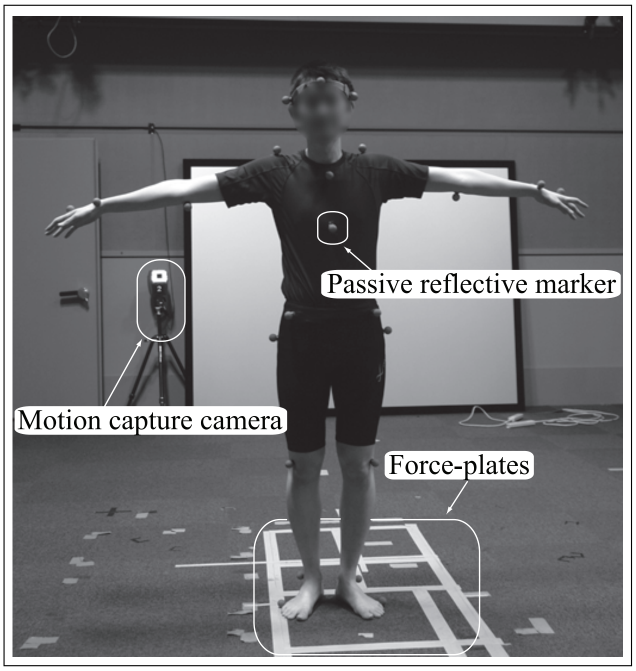


Fig. 11. Experimental setup of the behavior capture systems.

8.2. Human body in the air

This section shows the identification results of the human body using only aerial motion data. In order to analyze various human motions, a human musculoskeletal structure is often modeled and approximated as a rigid-body system because of its practicality (Delp and Loan, 1995; Yamane and Nakamura, 2000; Nakamura et al., 2005). Although the simulation of deformable objects can deal with the volumetric effects of human bodies (Teran et al., 2005), it suffers from the huge computational cost and the many model parameters to be measured or identified. In this paper, we consider a model of the human body made of 34 DOFs and 15 rigid links as described in Table 3. It represents the most important DOFs that are used in daily activities such as locomotion. In addition to the 34 DOFs of its kinematics, 6 DOFs are used to define the generalized coordinates corresponding to the base-link.

The composition matrix $\boldsymbol{X}_{j,j}$ of the link connected by a spherical joint can be chosen as follows:

$$\boldsymbol{X}_{j,j} = \begin{bmatrix} \mathbf{0}_9 & \boldsymbol{E}_9 \end{bmatrix}^{\mathrm{T}} \tag{126}$$

From Equations (38), (39), (40), (125), and (126), this kinematic model leads to 128 base parameters. The model can satisfy all of the conditions shown in Section 5, and thus the normalized base parameters are structurally identifiable. The numerical identifiability was also checked with the dynamics simulation.

The motion was recorded by the motion capture system as referred above. The experimental setup of the capture systems is shown in Figure 11. A total of 35 reflective optical markers attached to the body were captured by the system. These markers are modified Helen Hayes Hospital marker set (Kadaba et al., 1990). They are located at the defined anatomical rigid points to diminish the influence from non-rigid skin and muscle movement to insure accuracy of inverse kinematics computations. The geometric parameters of the model were measured by the position of the markers that were captured when the subject stands still. The regressor $\boldsymbol{Y}_{all}$ was computed from the position of markers in the same way mentioned in the identification of the two-link system. The contact forces were simultaneously measured by the two force plates already introduced in the previous section. This force information is not used for identification, but both to detect the moment in the air and validate the identification results.

For the identification of the unit vector $\widehat{\boldsymbol{\phi}}_{B,e}$, four sets of motion of the subject jumping on the force plates several times were recorded. The sequences in the air were then extracted from the data set. Some snapshots of the motion are shown in Figure 12. As the base parameter of the base-link representing the mass is equivalent to the total mass of the system, the standard base parameters $\widehat{\boldsymbol{\phi}}_B$ was also identified from (124) using both $\widehat{\boldsymbol{\phi}}_{B,e}$ and the mass of the subject $\widehat{M}$ measured by a scale.

Finally the obtained results are validated. For that, the measured contact forces $\boldsymbol{F}^{ext}$ (force-plate data) were compared with the contact forces that are computed with the identified parameters: $\widehat{\boldsymbol{F}}^{ext} \triangleq \boldsymbol{Y}_{OB}\widehat{\boldsymbol{\phi}}_B$. Another motion on the ground was also measured for the cross-validation. The motion itself has enough excitation to identify whole parameters, and seems like a kind of exercise of the whole body (Venture et al., 2009a).

The obtained results are presented in Figure 13. They give the comparison of the measured forces $\boldsymbol{F}^{ext}$ (gray thick lines), with the estimated forces $\widehat{\boldsymbol{F}}^{ext}$ (black thin line) obtained from the identified parameters, for each of the six components. The absolute mean error of six components is shown in the caption of Figure 13. From the cross-validation figures, the identified parameters allowed a good prediction of the generalized forces, and the mean errors were small with respect to the full scale.

9. Conclusion

In this paper we have studied the dynamics of legged systems in terms of the identification theory. In particular, we have focused on the under-actuated base characterizing the dynamics of legged systems. The important features of legged systems shown in this paper is summarized as follows.

A. The minimal set of inertial parameters appearing in the equation of motion of the under-actuated base of a legged system is equivalent to the set in the whole equations of the system. Therefore, the identification of base parameters of a legged system can be done only with the six base-link equations.

B. In particular, when a legged system has no external reaction force through the interaction with the environment, it can be regarded a kind of floating system under

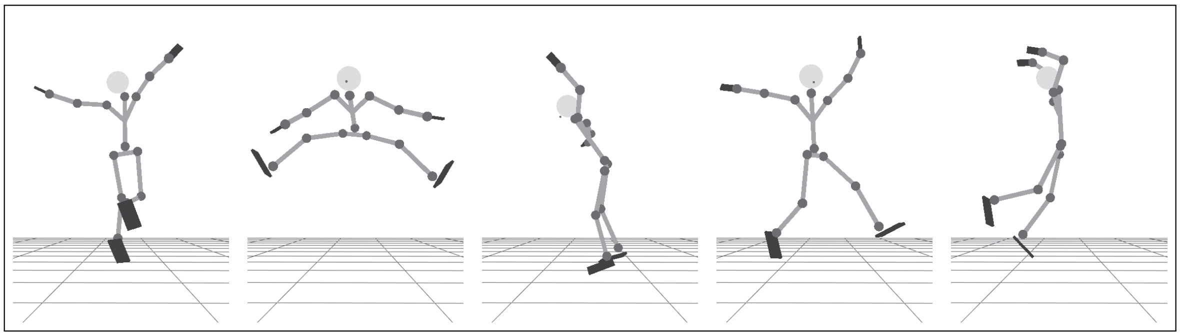


Fig. 12. Snapshots of the jumping motion used for identification.

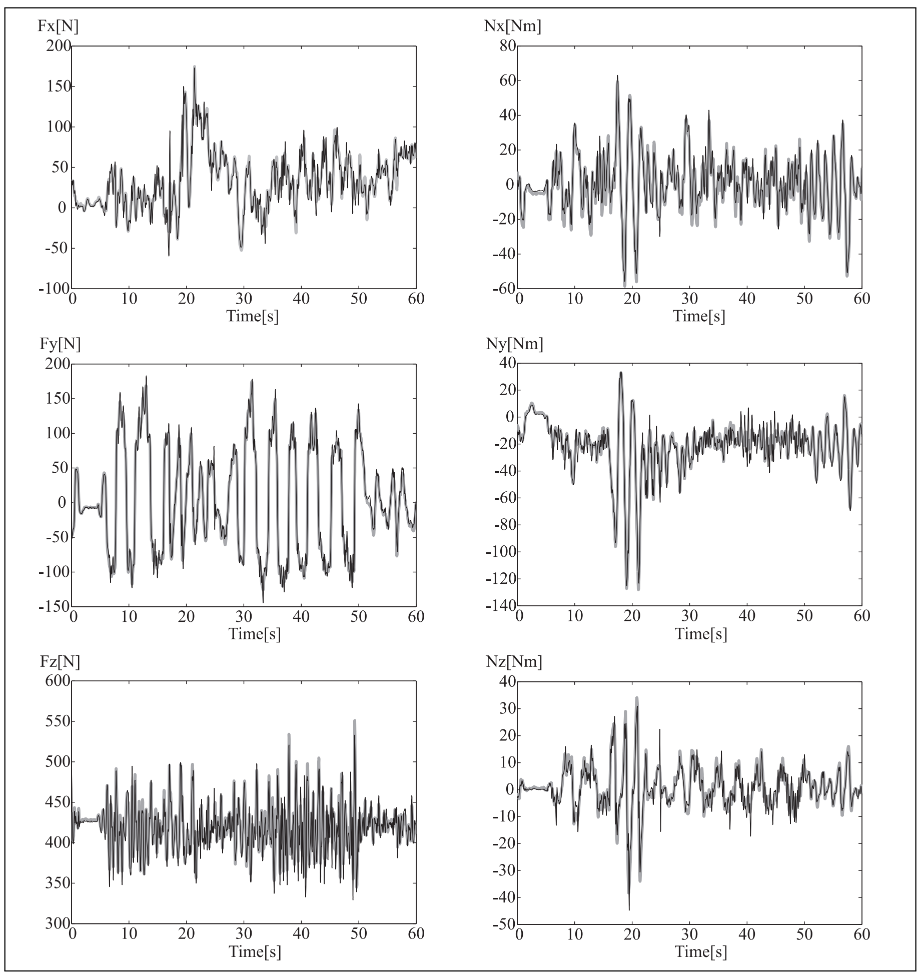


Fig. 13. Cross-validation of external forces for the motion exciting the whole body from aerial identification result. The gray thick lines show the measured forces, and the black thin lines show the estimated forces from the identified parameters. The mean errors of six-axis forces are as follows: Fx 6.38[N], Fy 5.90[N], Fz 7.08[N], Nx 2.86[Nm], Ny 3.47[Nm], Nz 2.12[Nm].

the law of conservation of angular momentum. The structurally unidentifiable parameters from the floating base-link dynamics is only the total mass, except for some special cases where the system consists of only one rigid body. Hence, all of the base parameters normalized by the total mass can be identified only from the generalized coordinates.

Based on the study, the practical methods to analyze the dynamics of legged systems such as humanoids and humans were presented. The features of the method are listed as follows:

1. The torque of each joint need not to be measured, and the measurement of the contact forces and the generalized coordinates is required.
2. The non-linear internal components such as friction need not to be considered, which means that the method can identify only inertial parameters without their influence.

3. As only the six equations of the base-link are used, the computational cost is quite low, which has a computational advantage in real-time identification or adaptive control.
4. Especially when the system has no external reaction force and under the law of conservation of momentum, the method does not require the measurement of forces.

The implementation method using force sensors was tested on a small-size humanoid robot. The seven different types of motions were recorded, and, for identification, the optimal set of motions was chosen among them according to the condition number of the combined regressor. The parameters were successfully identified from the set. The validations, using the two motions that are not used during the identification procedure, showed a good correlation between the external forces reconstructed from the parameters and those measured by the force plates; on the other hand, the forces estimated from the CAD data of the robot had large errors. With the results, the validity of the method and the importance of the identification step were shown.

The implementation method without force sensors was applied for two floating systems. First, the method was verified by experiments with a simple two-link system connected with a single rotational joint. The identified parameters were close to the measured parameters, and showed the validity of the proposed method. The method was also applied to identify the inertial parameters of the human body. Several jumping motions were recorded for identification, and one motion on the ground with enough excitation was also recorded for cross-validation. For the validation, the forces estimated from identified parameters during the motion on the ground were compared with the measured forces. The parameters allowed a good prediction of the measured forces with small errors.

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Appendix A: Proof of Lemma 1

According to Equations (17) and (19), we have $\boldsymbol{Y}_{O0} = {}^0\boldsymbol{Y}_0$. When $j = 0$, we also have $n_{J,0} = 6$, ${}^0\dot{\boldsymbol{p}}_0 = {}^0\dot{\boldsymbol{p}}_0^w$, ${}^0\ddot{\boldsymbol{p}}_0 = {}^0\ddot{\boldsymbol{p}}_0^w$, ${}^0\boldsymbol{\omega}_0 = {}^0\boldsymbol{\omega}_0^w$, ${}^0\dot{\boldsymbol{\omega}}_0 = {}^0\dot{\boldsymbol{\omega}}_0^w$, and $\boldsymbol{K}_0 = \boldsymbol{E}$. By using Equation (14), $\boldsymbol{Y}_{O0}$ is modified as

$$\boldsymbol{Y}_{O0} = \sum_{k=1}^{6} \boldsymbol{Y}_m(\boldsymbol{e}_{p0,k},\ \boldsymbol{e}_{q0,k})\,\ddot{q}_{0,k} + \sum_{k=1}^{6}\sum_{l=1}^{6} \begin{bmatrix} \boldsymbol{0}\ [\boldsymbol{e}_{q0,k}\times][\boldsymbol{e}_{q0,l}\times] & \boldsymbol{O} \\ \boldsymbol{0}\quad \boldsymbol{O} & [\boldsymbol{e}_{q0,k}\times][\boldsymbol{e}_{q0,l}\bullet] \end{bmatrix} \dot{q}_{0,k}\dot{q}_{0,l} \tag{127}$$

Let us define the following matrix:

$$\boldsymbol{Y}_{m0} \triangleq \left[\, \boldsymbol{Y}_m(\boldsymbol{e}_{p0,1},\ \boldsymbol{e}_{q0,1})^{\mathrm{T}} \cdots \boldsymbol{Y}_m(\boldsymbol{e}_{p0,6},\ \boldsymbol{e}_{q0,6})^{\mathrm{T}} \,\right]^{\mathrm{T}} \tag{128}$$

From $\boldsymbol{K}_0 = \boldsymbol{E}$ and Equation (14), we obtain

$$\boldsymbol{Y}_{m0}{}^{\mathrm{T}}\boldsymbol{Y}_{m0} = \mathrm{diag}(\begin{bmatrix} 3 & 4 & 4 & 4 & 1 & 1 & 1 & 2 & 2 & 2 \end{bmatrix}^{\mathrm{T}}) \quad (129)$$

As the number of rows of $\boldsymbol{Y}_{m0}$ is larger than that of its columns, rank($\boldsymbol{Y}_{m0}$) = 10 holds. By variation of only acceleration of the base-link, dim V($\boldsymbol{Y}_{O0}$) = rank($\boldsymbol{Y}_{m0}$) = 10 holds.

Appendix B: Proof of Lemmas 2 and 2'

Let us prove Lemmas 2 and 2$'$ at the same time under Conditions A, B, and C.

The matrix ${}^j\boldsymbol{Y}_j^{p_j}$ defined by (29) can be transformed as follows:

$$\begin{aligned} {}^j\boldsymbol{Y}_j^{p_j} &= \sum_{k=1}^{n_j} \boldsymbol{Y}_m({}^j\dot{\boldsymbol{p}}_{p_j}, {}^j\boldsymbol{\omega}_{p_j})\,\boldsymbol{B}_{Kj,k}\dot{q}_{j,k} \\ &+ \sum_{k=1}^{n_j}(\,\dot{q}_{j,k}{}^j\boldsymbol{A}_{dj} + \ddot{q}_{j,k}\boldsymbol{E}_6)\,\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k},\ \boldsymbol{e}_{qj,k}) \\ &+ \sum_{k=1}^{n_j}\sum_{l=1}^{n_j} \dot{q}_{j,k}\dot{q}_{j,l}\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k} \times \boldsymbol{e}_{qj,l},\ \boldsymbol{0}) \end{aligned} \quad (130)$$

There exists $\boldsymbol{x}_j^{\perp}$ satisfying Equation (48) for all $\dot{\boldsymbol{q}}_j, \ddot{\boldsymbol{q}}_j$. Therefore, it is necessary that $\forall k, l\ (1 \le k \le n_j,\ k < l \le n_j)$, $\boldsymbol{x}_j^{\perp}$ also satisfies (48) both when $\dot{q}_{j,k} = 0,\ \ddot{q}_{j,k} \ne 0$, and when $\dot{q}_{j,k} \ne 0,\ \dot{q}_{j,l} = 0,\ \ddot{q}_{j,k} = 0\ (l \ne k)$. Thus, from (130), it is necessary that $\boldsymbol{x}_j^{\perp}$ satisfies all three equations as follows:

$$\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k},\ \boldsymbol{e}_{qj,k})\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \quad (131)$$

$$\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k} \times \boldsymbol{e}_{qj,l},\ \boldsymbol{0})\boldsymbol{x}_{\mathbf{j}}^{\perp} = \boldsymbol{0} \quad (132)$$

$$[({}^j\boldsymbol{R}_{p_j}{}^{p_j}\boldsymbol{\omega}_{p_j})\,\bullet]\begin{bmatrix}\boldsymbol{0} & \boldsymbol{B}_C(\,\boldsymbol{e}_{pj,k}) & \boldsymbol{B}_W(\,\boldsymbol{e}_{qj,k})\end{bmatrix}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \quad (133)$$

Since there exist at least one $\boldsymbol{q}_c$ such that all of the elements of ${}^0\boldsymbol{w}_0$ and $\dot{\boldsymbol{q}}_c$ are linearly independent when $\boldsymbol{q}_c$ is fixed and $\dot{\boldsymbol{q}}_c \ne \boldsymbol{0}$, all three elements of ${}^j\boldsymbol{\omega}_{p_j}(= {}^j\boldsymbol{R}_{p_j}{}^{p_j}\boldsymbol{\omega}_{p_j})$ are linearly independent. Therefore, as can be seen from Equation (13), it is clear that dim V($[({}^j\boldsymbol{\omega}_{p_j})\,\bullet]$) = 6. Then, Equation (133) can be replaced with Equation (134).

$$\begin{bmatrix}\boldsymbol{0} & \boldsymbol{B}_C(\,\boldsymbol{e}_{pj,k}) & \boldsymbol{B}_W(\,\boldsymbol{e}_{qj,k})\end{bmatrix}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \quad (134)$$

If $\boldsymbol{x}_j^{\perp}$ satisfies Equations (131), (132), and (134), $\boldsymbol{x}_j^{\perp}$ also satisfies Equation (48). As Equations (131), (132), and (134) are equivalent with Equations (41) and (42), the necessary and sufficient condition that there exists $\boldsymbol{x}_j^{\perp}$ satisfying Equation (48) is that there exists $\boldsymbol{x}_j^{\perp}$ satisfying Equations (41) and (42).

Appendix C: Proof of Lemma 3

Here ${}^{p_j}\boldsymbol{B}_j$ is the transformation matrix used to express the inertial parameters. It is made of translational vector ${}^{p_j}\boldsymbol{p}_j$ and rotation matrix ${}^{p_j}\boldsymbol{R}_j$. In mechanics, they are second-order differentiable. The derivative of ${}^{p_j}\boldsymbol{B}_j$ can be computed as

$$\frac{d}{dt}{}^{p_j}\boldsymbol{B}_j = {}^{p_j}\boldsymbol{B}_j{}^j\boldsymbol{B}_{dj}^{p_j} \quad (135)$$

Since ${}^{p_j}\boldsymbol{B}_j$ is a regular matrix, Then a necessary and sufficient condition to have Equation (49) is that

$${}^j\boldsymbol{B}_{dj}^{p_j}\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \quad (136)$$

By using Equations (14) and (32), Equation (136) is modified as

$$\begin{aligned} &\sum_{k=1}^{n_j}(\,\dot{q}_{j,k}{}^j\boldsymbol{A}_{dj} + \ddot{q}_{j,k}\boldsymbol{E})\,\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k},\ \boldsymbol{e}_{qj,k})\boldsymbol{x}_j^{\perp} \\ &\quad + \sum_{k=1}^{n_j}\sum_{l=1}^{n_j}\dot{q}_{j,k}\dot{q}_{j,l}\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k} \times \boldsymbol{e}_{qj,l},\ \boldsymbol{0})\boldsymbol{x}_{\mathbf{j}}^{\perp} = \boldsymbol{0} \end{aligned} \quad (137)$$

If Equations (41) and (42) hold, then

$$\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k},\ \boldsymbol{e}_{qj,k})\boldsymbol{x}_j^{\perp} = \boldsymbol{0} \quad (138)$$

$$\boldsymbol{Y}_m(\,\boldsymbol{e}_{pj,k} \times \boldsymbol{e}_{qj,l},\ \boldsymbol{0})\boldsymbol{x}_{\mathbf{j}}^{\perp} = \boldsymbol{0} \quad (139)$$

Therefore, if Equations (41) and (42) hold, then Equations (136) and (49) hold.